An

Inter Disciplinary Mini Project
Report on

"Part A: Phytochemical Synthesis of Semiconducting Copper Oxide (CuO) Using Mulberry Extract in Ethanol by Spray Pyrolysis Method

&

Part B: GreenSynth: A Data-Driven Research and Analytics Platform for Green Synthesis of Semiconductor Materials"

SUBMITTED TO THE PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY, SOLAPUR IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF

BACHELOR OF TECHNOLOGY DEGREE IN COMPUTER SCIENCE AND ENGINEERING

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UNDER THE GUIDANCE OF

Dr. S. H. Pawar

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
N. B. NAVALE SINHGAD COLLEGE OF ENGINEERING,
SOLAPUR - 413 255

2026-27

AFFILIATED TO

PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY, SOLAPUR

SPSPM
॥ ज्ञान दान पवित्र दान ॥

CERTIFICATE

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is a bonafide work carried out by above students under the supervision of Dr. S. H. Pawar and it is submitted towards the partial fulfillment of the requirement of Punyashlok Ahilyadevi Holkar Solapur University, Solapur for the award of the degree of Bachelor in Technology (Computer Science and Engineering) during academic year 2026-27.

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DECLARATION

We hereby declare that the work embodied in this Project Report Entitled **"Part A: Phytochemical Synthesis of Semiconducting Copper Oxide (CuO) Using Mulberry Extract in Ethanol by Spray Pyrolysis Method & Part B: GreenSynth: A Data-Driven Research and Analytics Platform for Green Synthesis of Semiconductor Materials"** is carried out by us in partial fulfillment of the degree Bachelor in Technology (Computer Science and Engineering) from N. B. Navale Sinhgad College of Engineering, Solapur and we have not submitted the same to any other University/Institute for the award of any other degree.

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ABSTRACT

The synthesis and optimization of functional semiconductor materials increasingly demand the integration of reproducible wet-laboratory protocols with automated computational data management and analytics. This Inter-Disciplinary Mini Project (IDMP) presents a dual-track investigation combining the green, phytochemical-assisted synthesis of semiconducting copper oxide (CuO) thin films with the development of **GreenSynth Analytics**, a specialized research data management and materials informatics platform.

In the experimental track (Part A), nanocrystalline CuO thin films are prepared through a phytochemical route utilizing mulberry (*Morus alba*) leaf extract in ethanol, with copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) and sodium hydroxide (NaOH) specified as the copper precursor system. Thin films are deposited onto preheated glass substrates using the chemical spray pyrolysis technique. The resulting films are evaluated through multi-technique characterization, including X-ray diffraction (XRD) for crystal structure, ultraviolet-visible (UV-Vis) spectroscopy for optical absorption and band gap determination, Fourier-transform infrared (FTIR) spectroscopy for chemical bonding, scanning electron microscopy (SEM) for surface morphology, and current-voltage (I–V) measurements for electrical transport properties.

In the software track (Part B), GreenSynth Analytics provides a structured, full-stack digital environment designed to address data fragmentation, manual transcription errors, and observer subjectivity in materials research workflows. Built with FastAPI and scientific computing libraries, the platform features tamper-evident file ingestion with cryptographic SHA-256 verification, deterministic scientific calculation engines (automated Scherrer crystallite sizing, maximum first-derivative Tauc optical band gap extraction, and Ohmic transport regression), machine-learning regression benchmarking, and automated ReportLab PDF research dossier generation. A rigorous distinction is maintained between measured laboratory observations and physics-grounded synthetic benchmark datasets used for software testing. Together, the two tracks establish a traceable, closed-loop research workflow connecting laboratory materials synthesis with automated scientific computing.

Keywords: Copper Oxide (CuO), Spray Pyrolysis, Phytochemical Synthesis, Mulberry Extract, Copper Sulfate Pentahydrate, Sodium Hydroxide, Materials Informatics, GreenSynth Analytics, Thin Film Characterization, Data Provenance.

Project Report Outline

The report is organized chronologically in terms of the objectives accomplished:

- **Chapter One (Introduction):** Introduction to phytochemical CuO synthesis, spray pyrolysis, GreenSynth Analytics, problem statement, scope, objectives, and interdisciplinary relevance.
- **Chapter Two (Literature Review):** Review of green synthesis of metal oxides, spray pyrolysis deposition physics, materials informatics platforms, and research gap identification.
- **Chapter Three (System Requirements & Setup):** Laboratory chemicals (copper acetate monohydrate, ethanol, mulberry extract), experimental apparatus (Soxhlet, spray pyrolysis setup, instruments), software requirements, and technology stack.
- **Chapter Four (System Design):** Architectural design, mathematical formulations (Scherrer, Tauc, Ohmic regression), UML diagrams, database schema, and process flowcharts.
- **Chapter Five (Methodology & Implementation):** Detailed extraction protocol, precursor preparation, spray pyrolysis deposition parameters, characterization procedures, and software module implementation.
- **Chapter Six (Testing & Validation):** Software verification (118 automated tests), SHA-256 cryptographic integrity tests, and laboratory instrument calibration procedures.
- **Chapter Seven (Results & Discussion):** Experimental characterization observations (XRD, UV-Vis, FTIR, SEM, I–V), automated property calculations, and machine-learning benchmarking results.
- **Chapter Eight (Conclusion & Future Scope):** Summary of findings, software evaluation, and future research directions.
- **Appendix A:** List of abbreviations used in the report.
- **Appendix B:** List of publications and project artifacts.
- **Appendix C:** Bibliography and references.

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Chapter 1

INTRODUCTION

1.1 Introduction to Project

Scientific research increasingly involves the integration of experimental investigation with computational tools for data management, analysis, and interpretation. In materials science, the preparation of a material is only one part of the research process. Understanding its structural, optical, morphological, and electrical properties is equally important for evaluating its potential applications. The systematic recording of synthesis conditions and characterization results is therefore essential for meaningful comparisons, reproducibility, and further investigation.

The present project integrates two complementary components: the phytochemical-assisted synthesis of semiconducting copper oxide (CuO) thin films using the spray pyrolysis technique, and the development of **GreenSynth Analytics**, a research data management and materials analysis platform. The experimental component focuses on the preparation and characterization of CuO thin films using mulberry extract and ethanol, while the software component provides a structured environment for organizing experimental information, managing characterization data, performing scientific calculations, and exploring relationships between synthesis parameters and material properties.

The project follows an interdisciplinary approach in which laboratory-based materials research is supported by software engineering, scientific computing, data visualization, statistical analysis, and machine-learning techniques. The experimental investigation provides the scientific context, while GreenSynth Analytics is intended to improve the organization, traceability, and interpretation of research data.

1.1.1 Part A: Phytochemical Synthesis and Spray Pyrolysis of CuO Thin Films

Copper oxide (CuO) is a semiconducting metal oxide that has attracted research interest because of its optical, electrical, and structural properties. Its potential applications include sensors, photovoltaic devices, catalytic systems, and other electronic and optoelectronic technologies. The properties of CuO depend on factors such as its preparation method, composition, microstructure, substrate conditions, and subsequent processing.

Conventional material synthesis can involve different chemical precursors, solvents, and processing conditions. Phytochemical-assisted synthesis investigates the use of plant-derived extracts as part of the preparation process. Such extracts contain naturally occurring chemical constituents that may influence precursor chemistry, nucleation, growth, or surface characteristics. However, their precise role depends on the extract composition and experimental conditions and must be investigated using appropriate evidence.

In this project, mulberry extract and ethanol are used in the preparation process, with copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) and sodium hydroxide (NaOH) in a

1:2 molar ratio specified as the copper precursor system. The spray pyrolysis technique is employed for the deposition of CuO thin films. In this technique, a precursor solution is atomized into small droplets and transported towards a heated substrate. The droplets undergo solvent evaporation and thermal transformations, contributing to the formation of a deposited film. The resulting film characteristics depend on the solution composition and process parameters, including substrate temperature, spray rate, spray duration, nozzle-to-substrate distance, and other relevant conditions.

The experimental investigation focuses on the relationship between synthesis conditions and the properties of the resulting films. Characterization methods such as X-ray diffraction (XRD), ultraviolet-visible (UV-Vis) spectroscopy, Fourier-transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), and electrical current-voltage (I–V) measurements can provide information about structural, optical, chemical-bonding, morphological, and electrical characteristics, respectively.

The scientific component therefore aims to establish a systematic experimental procedure, record the synthesis conditions, characterize the prepared films using the methods actually performed, and interpret the resulting observations. The findings are intended to contribute to a better understanding of the relationship between preparation conditions and the properties of CuO thin films.

1.1.2 Part B: GreenSynth Analytics

Experimental materials research generates different forms of information, including synthesis parameters, sample identifiers, laboratory observations, characterization files, processed datasets, and calculated material properties. When this information is maintained in separate spreadsheets, files, or scripts, it can become difficult to trace results back to their original experiments, compare samples consistently, and reproduce analytical procedures.

To address these challenges, GreenSynth Analytics has been developed as a research data management and materials analysis platform. Its purpose is to provide a structured digital environment for managing research projects, experiments, synthesis parameters, samples, characterization records, raw files, and derived scientific properties.

The platform connects experimental records with subsequent data-processing and analytical workflows. Depending on the implemented functionality, these workflows can include data validation, scientific calculations, graphical visualization, statistical analysis, machine-learning model evaluation, and design-of-experiments support. Organizing these activities within a common system can help researchers investigate patterns in experimental data and compare different synthesis conditions more systematically.

GreenSynth Analytics also emphasizes data traceability by maintaining relationships between experiments, samples, characterization records, and calculated outputs. This structure supports the identification of the data and processing steps associated with a reported result. Visualization and analytical reports can further help users interpret complex datasets and communicate findings more effectively.

The platform is designed around the needs of materials research, with the CuO synthesis investigation serving as a principal project context. Its architecture can potentially be extended to additional materials and synthesis techniques by introducing appropriate experimental parameters, characterization workflows, and property calculations.

An important distinction is maintained between measured laboratory data and synthetic data generated for software testing or analytical pipeline evaluation. Synthetic data can be useful for checking data integrity, testing software functionality, and benchmarking computational workflows, but it cannot establish the experimentally measured properties of CuO films. Scientific conclusions must therefore be based on verified experimental measurements and appropriate analysis.

Table 1.1 summarizes the comparative dimensions of the two co-designed tracks.

Table 1.1: Multi-Dimensional Comparison of Hardware and Software Project Tracks

Dimension	**Part A: Hardware (Physical Science)**	**Part B: Software (Materials Informatics)**
Core Discipline	Materials Science / Nanotechnology	Computer Science / Informatics
Primary Objective	Phytochemical synthesis of CuO thin films via spray pyrolysis	Automated ingestion, physics derivation, ML benchmarking, and PDF reporting
Input Entities	*Morus alba* leaf extract, ethanol, copper acetate monohydrate, glass substrates	Raw instrumental ASCII/CSV files (XRD, UV-Vis, FTIR, I–V), synthesis logs
Processing Tools	Soxhlet extractor, magnetic stirrer, centrifuge, spray pyrolysis unit, heating stage	FastAPI, SQLAlchemy ORM, SQLite/PostgreSQL, Scikit-learn, React/TypeScript
Analytical Output	Physical nanocrystalline thin films, optical band gap, electrical conductivity	Deterministic physical metrics (D, E_g, σ), ML leaderboard, PDF dossier
Verification Method	Calibrated instruments (pH meter, spectrophotometer, conductivity meter)	118 automated tests (`pytest`), SHA-256 hashes, 5-fold cross-validation

1.2 Problem Statement

Materials research involves multiple connected activities, beginning with the preparation of a material and continuing through characterization, data processing, property calculation, comparison, and interpretation. The reliability of the resulting conclusions depends on the quality of the experimental procedure, the accuracy of the measurements, and the traceability of the data used in the analysis.

In phytochemical-assisted synthesis of CuO thin films, several experimental conditions may influence the resulting material properties. Understanding these relationships requires the systematic recording of preparation parameters and the analysis of characterization results. Differences in sample preparation, measurement conditions, data-processing procedures, or record-keeping practices can complicate comparisons between experiments. Furthermore, the contribution of plant-derived constituents to the synthesis process cannot be established solely from the use of a plant extract; it requires appropriate experimental and characterization evidence.

A related challenge is the management of the data generated during the research process. Synthesis records, sample details, raw characterization files, calculated prop-

erties, and analytical results may exist in different formats and locations. Without a structured system, connecting a reported property to its original sample, experimental conditions, raw data, and calculation procedure can become difficult. Repeated manual processing may also introduce inconsistencies and reduce the efficiency of data comparison.

The project therefore addresses two interconnected problems:

1. **Scientific Problem:** To investigate the preparation and characterization of phytochemical-assisted CuO thin films and examine how documented synthesis conditions relate to their structural, optical, morphological, and electrical properties.

2. **Software Problem:** To develop a structured research platform that organizes experimental data, maintains relationships between experiments and samples, supports scientific data processing and visualization, and provides analytical functionality for systematic comparison and investigation.

GreenSynth Analytics is intended to address the software-related challenges while supporting the broader scientific investigation. The success of the software must be assessed through implementation and testing evidence, whereas conclusions about the synthesized material must be supported independently by verified laboratory data.

1.3 Scope and Objectives

The project combines experimental materials science with the development of a scientific software platform. Its scope includes the preparation and investigation of CuO thin films, the management of associated research data, and the implementation and evaluation of relevant analytical workflows.

1.3.1 Scope

A. Scientific and Experimental Scope

- Preparation of mulberry extract using the documented laboratory extraction procedure, including Soxhlet extraction where performed.
- Preparation of the precursor solution using copper acetate monohydrate and ethanol according to the recorded experimental procedure.
- Deposition of CuO thin films using spray pyrolysis.
- Recording of relevant synthesis parameters and experimental observations.
- Characterization of the prepared films using the techniques actually performed, including XRD, UV-Vis, FTIR, SEM, and electrical I–V measurements where data are available.
- Processing and interpretation of characterization data using appropriate scientific methods.

- Investigation of relationships between synthesis conditions and material properties, subject to the availability and quality of experimental measurements.

B. Software Development Scope

- Structured management of research projects, experiments, synthesis parameters, and samples.
- Storage and organization of characterization records and raw research files.
- Association of raw data with relevant experiments and samples.
- Support for scientific calculations and derived-property records.
- Visualization and comparison of research data.
- Statistical analysis and machine-learning workflows, where implemented.
- Evaluation of predictive models using appropriate validation methods.
- Design-of-experiments and candidate-condition analysis, where implemented.
- Data integrity, access control, validation, and testing of relevant software functions.
- Preparation of research outputs and analytical reports for interpretation and communication.

C. Scope Limitations

The project is limited to the materials, synthesis methods, characterization techniques, and software functions included in the actual implementation and experimental work. Additional materials, instruments, synthesis methods, or analytical features are outside the demonstrated scope unless separately implemented and evaluated.

Machine-learning outputs and proposed experimental conditions are treated as analytical support rather than definitive scientific conclusions. Their reliability depends on the dataset, experimental coverage, model-validation procedure, and applicability to the conditions being investigated. Synthetic datasets used for software testing are kept distinct from experimental measurements.

1.3.2 Objectives

A. Scientific Objectives

1. To investigate the phytochemical-assisted synthesis of semiconducting CuO thin films using mulberry extract and the spray pyrolysis technique.
2. To prepare the precursor solution using the specified copper precursor and solvent according to the documented experimental procedure.
3. To identify and record the synthesis parameters relevant to the preparation of CuO thin films.
4. To characterize the prepared films using the available structural, optical, chemical, morphological, and electrical characterization techniques.

5. To process and interpret the characterization results using appropriate scientific calculations and analytical methods.
6. To investigate relationships between synthesis conditions and the observed material properties, where supported by experimental evidence.

B. Software Objectives

1. To design and develop GreenSynth Analytics as a structured platform for materials research data management and analysis.
2. To organize research projects, experiments, synthesis parameters, samples, characterization records, and calculated properties.
3. To establish traceable relationships between raw research files, experimental records, and derived analytical outputs.
4. To implement data validation, processing, scientific calculations, and visualization workflows according to the platform requirements.
5. To support statistical analysis and machine-learning model evaluation where the necessary functionality is implemented.
6. To investigate the use of design-of-experiments methods for comparing synthesis conditions and identifying candidate experiments.
7. To distinguish synthetic test data from laboratory measurements and maintain appropriate interpretation of analytical outputs.

C. Integrated Objectives

1. To connect the scientific workflow of CuO synthesis and characterization with a structured digital research-data workflow.
2. To improve the traceability and organization of experimental information.
3. To support systematic comparison and interpretation of synthesis conditions and material properties.
4. To establish a foundation for extending the platform to additional materials and research projects.

1.4 Identification of Need

The need for the present project arises from the combination of experimental complexity in materials synthesis and the growing importance of structured research-data management:

1. **Process Parameter Sensitivity in Thin Film Deposition:** The preparation of CuO thin films involves multiple process variables that may influence the resulting material. Recording these conditions consistently is necessary to understand experimental variation and compare results. A systematic workflow helps ensure that the interpretation of a material property considers the conditions under which the corresponding sample was prepared.

2. **Multi-Modal Characterization Data Streams:** Characterization produces data in different forms. XRD and spectroscopic measurements produce numerical datasets and plots, SEM produces images, and electrical measurements produce current-voltage data. These different formats require appropriate processing and interpretation. Maintaining their relationship with sample identities and synthesis conditions is important for traceability.

3. **Scientific Derivation Traceability:** Scientific calculations require explicit methods, assumptions, units, and input data. A structured system can help preserve the relationship between raw measurements, processing steps, and derived properties. This is particularly important when results are compared across samples or used in subsequent statistical analyses.

4. **Scalable Research Data Management:** Research datasets can become increasingly difficult to manage as the number of experiments, samples, and characterization records grows. A dedicated research platform can reduce dependence on disconnected files and provide a more consistent way to organize and retrieve information.

5. **Data-Driven Pattern Discovery and Screening:** Data visualization and statistical analysis can help researchers explore patterns, identify unusual observations, and compare experimental conditions. Predictive models may provide additional analytical support when sufficient suitable data are available. However, these techniques do not eliminate the need for sound experimental design or independent laboratory validation.

6. **Integrated Digital Workflow for Materials Science:** A platform designed specifically around materials research can connect the experimental workflow with data processing and reporting. GreenSynth Analytics addresses this need by providing a common digital environment for research records and supported analytical workflows. Its effectiveness depends on the correctness of the implementation, the quality of the data, and appropriate testing.

1.5 Motivation

The motivation for this project is to combine practical materials research with software-based methods for organizing and interpreting scientific information.

From the **scientific perspective**, CuO thin films provide a context for investigating the relationship between synthesis conditions and material properties. The use of mulberry extract introduces a phytochemical-assisted aspect to the preparation process, while spray pyrolysis provides a defined deposition technique. Studying the resulting films can help develop an understanding of how experimental conditions relate to the properties observed through characterization.

From the **software engineering perspective**, the project is motivated by the need for a structured, traceable, and extensible environment for materials research. A research platform should do more than store isolated measurements. It should maintain connections between experiments, samples, raw files, processing operations, calculated properties, and analytical outputs so that the origin of a result can be examined.

Another motivation is the opportunity to apply **scientific computing and data analysis** to materials research. Appropriate visualization, statistical methods, and machine-learning techniques can help researchers investigate relationships in available datasets and compare candidate experimental conditions. The use of such methods also creates an opportunity to study the limitations of data-driven analysis, including insufficient data, uncertainty, overfitting, and differences between synthetic and experimentally measured values.

The development of GreenSynth Analytics provides practical experience in integrating frontend development, backend services, database design, scientific computation, data visualization, and research workflows. At the same time, the CuO synthesis investigation provides a scientific context in which the relevance and limitations of these software capabilities can be examined.

The overall motivation is therefore to establish a connection between laboratory experimentation and structured computational analysis, supporting more systematic research while preserving the distinction between experimental evidence and computational predictions.

1.6 Relevance to Interdisciplinary Approach

The present project is interdisciplinary because it combines materials science, chemistry, laboratory experimentation, software engineering, scientific computing, and data analytics within a common research workflow:

- **Materials Science and Semiconductor Physics:** These disciplines provide the basis for understanding CuO, its relevant properties, and the relationship between material structure, morphology, optical response, and electrical behavior.
- **Chemistry and Phytochemical-Assisted Synthesis:** Chemical principles are relevant to precursor preparation, solvent selection, extract preparation, and the interpretation of possible interactions between plant-derived constituents and the synthesis process. Any proposed chemical mechanism must be supported by appropriate experimental evidence and literature.
- **Laboratory Techniques and Instrumentation:** Soxhlet extraction, spray pyrolysis, sample preparation, and characterization procedures provide the experimental foundation of the project. The quality of the resulting data depends on instrument conditions, calibration where applicable, and consistent experimental practice.
- **Software Engineering:** GreenSynth Analytics applies requirements analysis, system architecture, database design, API development, frontend implementation, access control, testing, and software validation to the organization of research information.
- **Scientific Computing and Data Analysis:** Numerical processing, statistical analysis, and scientific calculations support the transformation of raw measurements into interpretable properties. Machine-learning methods may be used to investigate relationships and evaluate predictions when appropriate data and validation procedures are available.

- **Data Visualization and Reporting:** Graphical representations and interactive reports help communicate experimental conditions, characterization outputs, trends, and comparisons. Visualization tools, including those integrated into GreenSynth or connected reporting tools, can make complex datasets easier to explore without replacing the underlying scientific analysis.

The interdisciplinary relationship is reflected in the overall workflow: experimental conditions define the preparation process; characterization generates raw evidence; scientific processing derives relevant properties; GreenSynth organizes and connects the resulting records; and statistical or computational analysis supports further investigation. Laboratory validation remains necessary to establish whether predicted or proposed conditions produce the expected material properties.

Thus, the project brings together experimental investigation and software-based research support. Its interdisciplinary contribution lies in connecting these activities within a traceable workflow while recognizing that software correctness and scientific validity require separate forms of verification.

Table 1.2 illustrates the mapping between physical materials processes and their corresponding computational informatics modules.

Table 1.2: Interdisciplinary Co-Design Matrix: Physical Synthesis to Computational Informatics Mapping

Wet-Lab Physical Process	**Physical Variables / Instrument**	**GreenSynth Software Module**	**Computational Output & Function**
Phytochemical Extraction	*Morus alba* mass, ethanol volume, Soxhlet reflux temp (80°C)	Experiment & Parameter Management Service	Immutable parameter snapshot, 3NF database logging
Precursor Preparation	Copper acetate monohydrate concentration, ethanol/extract volume ratio, stirring time	Precursor Solution Configuration Engine	Stoichiometry validation, solution property logging
Spray Pyrolysis Deposition	Substrate temp (200–450°C), spray rate, carrier pressure, nozzle distance	Synthesis Parameter Configuration Studio	Automated parameter range validation, unit conversion
X-Ray Diffraction (XRD)	Diffractometer (2θ–I arrays)	Automated XRD Scherrer Engine	Baseline subtraction, peak detection, crystallite size D
UV-Vis Spectroscopy	Spectrophotometer (190–1100 nm)	Automated UV-Vis Tauc Extrapolation Engine	Absorption coefficient α, maximum derivative tangent, band gap E_g
Electrical I–V Transport	Two-point probe source-meter (-2.0 V to $+2.0$ V)	Automated Electrical Transport Engine	Linear least-squares Ohmic regression, resistance R, conductivity σ
Closed-Loop Optimization	Next-experiment candidate selection	ML Cross-Validation & Applicability Engine	5-Fold CV model benchmarking, domain bounds, ReportLab PDF dossier

Chapter 2

LITERATURE REVIEW

2.1 Existing Systems

Existing research relevant to the present project can be divided into two major areas: phytochemical-assisted synthesis and characterization of copper oxide (CuO), and software platforms for scientific data management and materials informatics. Reviewing both areas establishes the scientific background for the experimental investigation and the technological context for GreenSynth Analytics.

2.1.1 Existing Approaches to Phytochemical-Assisted CuO Synthesis

Plant-mediated synthesis of metal and metal oxide nanomaterials has received considerable research attention because plant extracts contain diverse chemical constituents that may participate in material formation or influence particle growth and surface properties. Phenolic compounds, flavonoids, and other phytochemicals have been investigated for their possible roles in reducing, complexing, capping, or stabilizing species during synthesis. However, the specific role of these constituents depends on the plant source, extraction procedure, precursor chemistry, and reaction conditions.

Siddiqi and Husen (2020) reviewed plant-metabolite-based approaches for the fabrication of copper and copper oxide nanoparticles. Their review describes the use of different plant sources and discusses the relevance of phytochemicals to nanoparticle formation and stabilization. The diversity of plant extracts and synthesis conditions indicates that experimental results obtained using one botanical source cannot automatically be generalized to another.

A review by *New Frontiers in the Plant Extract Mediated Biosynthesis of Copper Oxide (CuO) Nanoparticles and Their Potential Applications* (2022) examined plant-extract-assisted approaches to CuO nanoparticle synthesis, factors influencing the process, and possible applications. The review also identified challenges associated with understanding synthesis mechanisms and achieving consistent material properties. These observations support the need to record experimental conditions carefully and interpret characterization results using evidence rather than assumptions.

The present project investigates a specific preparation route involving mulberry (*Morus alba*) extract, ethanol, copper acetate monohydrate, and spray pyrolysis. Its scientific scope concerns CuO thin films rather than assuming that findings for solution-synthesized nanoparticles necessarily apply to deposited films. The resulting material properties must therefore be evaluated using the actual preparation and characterization procedures employed in this project.

2.1.2 Existing Synthesis and Characterization Techniques

Different physical and chemical methods are used to prepare copper oxide materials, including solution-based synthesis, precipitation, hydrothermal processing, and deposition techniques. The choice of method influences the processing conditions, form of the resulting material, and suitability for the intended application.

Spray pyrolysis is a deposition technique in which an atomized precursor solution is transported towards a heated substrate. Solvent evaporation and thermal transformations contribute to film formation. The resulting properties can depend on precursor composition, substrate temperature, droplet transport, spray rate, deposition duration, and other operating conditions.

Material characterization is required to establish the properties of the prepared film. X-ray diffraction (XRD) is used to investigate crystal structure and phase information. Ultraviolet-visible (UV-Vis) spectroscopy provides optical measurements that may be used for band-gap estimation when appropriate assumptions and data transformations are applied. Fourier-transform infrared spectroscopy (FTIR) can provide evidence about vibrational modes and functional groups. Scanning electron microscopy (SEM) is used to examine surface morphology, while electrical current-voltage (I–V) measurements can support the evaluation of electrical behavior.

These techniques provide different and complementary forms of evidence. No single technique establishes every property of a material. For example, surface morphology observed through SEM does not by itself establish chemical composition, and a proposed phytochemical mechanism cannot be confirmed solely from the use of a plant extract.

Consequently, the scientific component of the present project requires a traceable connection between preparation conditions, sample identity, raw characterization data, processing procedures, and derived properties.

2.1.3 Existing Materials Data and Informatics Platforms

Materials research increasingly uses digital databases, scientific software, and computational methods to manage information and investigate relationships between material composition, processing conditions, structure, and properties.

Hill et al. (2016) discussed the growing importance of large-scale data and informatics in materials science. Their work highlighted challenges such as fragmented data resources, inconsistent data standards, and difficulties in making heterogeneous scientific information accessible and reusable. The study also described materials-data resources and the potential role of analytical platforms in accelerating materials research.

The Materials Project is an established open platform that provides access to computational materials data and related tools. It supports the exploration of material properties and provides programmatic access to data for further analysis. Its principal emphasis is on materials data and computationally derived properties rather than serving as a dedicated laboratory workflow for the specific CuO synthesis process investigated here.

Citrine Informatics and its Citrination platform represent another approach to materials data infrastructure and informatics. O'Mara, Meredig, and Michel (2016) examined the design of a materials data infrastructure, including the import, storage,

and access of structured and unstructured information. Such platforms illustrate the importance of connecting scientific records with interfaces and analytical tools.

The Materials Data Facility, described by Blaiszik et al. (2016), supports materials research data publication and discovery. Its approach includes metadata, programmatic interfaces, data discovery, and the sharing of research datasets. These functions demonstrate how structured metadata and consistent data access can improve the reuse and discoverability of scientific information.

Rumble (2017) discussed broader challenges in accessing materials data, including fragmentation across resources, interoperability, and data quality. The review emphasized the importance of findability, accessibility, interoperability, and reusability, commonly referred to as the FAIR principles.

These systems provide important precedents for scientific data management and materials informatics. However, their objectives and intended users differ. A general materials database, a computational property repository, a data-publishing infrastructure, and a project-specific experimental research platform do not necessarily provide identical functionality.

2.1.4 Existing Data Analysis and Visualization Approaches

Scientific data analysis commonly combines numerical processing, statistical methods, graphical visualization, and predictive modeling. Python-based scientific libraries provide tools for data transformation, numerical calculations, statistical analysis, and machine learning. Visualization tools can represent distributions, correlations, trends, and comparisons between experimental conditions.

Materials-informatics research demonstrates that structured data can be used to investigate process-property relationships and develop predictive models. However, the usefulness of such models depends on data quality, sample coverage, appropriate validation, and the distinction between measured and calculated quantities.

Data visualization platforms, including Power BI, can provide interactive reports, filtering, summaries, and comparisons. These tools are useful for communicating research information, but they do not replace specialized scientific processing. For example, a plotted band-gap value must still be derived using an appropriate scientific method, and an electrical property must still be calculated from valid measurements and geometry.

GreenSynth Analytics follows this broader approach by combining structured research records with scientific processing and visualization. Its specific contribution must be evaluated against the functions actually implemented rather than assumed from the capabilities of other platforms.

2.2 Limitations of Existing Systems

Although existing synthesis methods, scientific databases, and materials-informatics platforms provide valuable capabilities, several challenges remain relevant to the present project.

2.2.1 Limitations in Phytochemical-Assisted Synthesis

Plant-mediated synthesis involves natural extracts whose chemical composition can vary with botanical source, growing conditions, plant part, extraction method, and storage. These variations may influence synthesis outcomes and complicate comparisons between studies.

Other challenges include the need to control processing conditions, achieve reproducible material characteristics, and distinguish proposed chemical mechanisms from experimentally established ones. The use of a plant extract alone does not establish that a synthesis route is environmentally preferable in every respect; solvent use, energy consumption, waste generation, and processing requirements also need consideration.

In addition, studies involving nanoparticles cannot automatically be treated as direct evidence for thin-film deposition. Differences in material form and preparation technique affect the interpretation of the findings. The present project must therefore evaluate the specific mulberry-extract and spray-pyrolysis procedure used for its CuO thin films.

2.2.2 Limitations in Experimental Data Management

Research information can be distributed across spreadsheets, instrument-generated files, laboratory notebooks, and separate processing scripts. When these resources are not connected through consistent identifiers and metadata, it can be difficult to trace a calculated property to the original sample and experimental conditions.

Differences in file formats, units, parameter names, and data-processing procedures can also make direct comparisons difficult. Inadequate recording of processing steps may reduce reproducibility, while incomplete metadata may prevent researchers from determining how a result was produced.

Materials-data infrastructure research has identified fragmentation, interoperability, and data-quality challenges as important issues in scientific data access and reuse (Hill et al., 2016; Rumble, 2017).

2.2.3 Limitations in Existing Materials Informatics Platforms

Existing materials-data systems serve different purposes, and no single limitation applies equally to all of them. Public materials repositories may focus on curated or computationally generated properties, while data-publishing platforms emphasize discovery, sharing, and metadata. Specialized commercial systems may provide broader experimental-data and analytics functionality.

For the present project, the relevant design challenge is to support the complete workflow required by the specific research context: recording synthesis conditions, associating samples with characterization files, preserving processing information, storing derived properties, and comparing analytical outputs.

A general database or visualization tool may not independently provide all the domain-specific calculations and sample-to-experiment relationships required for this workflow. Such functionality may require additional configuration, integration, or custom development. This is a design consideration for GreenSynth Analytics, not a claim that existing platforms universally lack these capabilities.

2.2.4 Limitations in Data-Driven Analysis

Statistical analysis and machine-learning models can help identify relationships in experimental data, but their conclusions depend on the quantity, quality, and representativeness of the available records. Small datasets, inconsistent experimental procedures, missing values, and duplicated or related samples can lead to misleading evaluation results.

A model may also perform well on data similar to its training records while failing to generalize to new experimental conditions. Evaluation should therefore use appropriate validation procedures, relevant baseline models, and metrics suited to the prediction task.

Furthermore, correlations between synthesis parameters and material properties do not independently establish causation. Predictions generated from synthetic data can help test a software pipeline but cannot validate the physical behavior of an actual CuO synthesis process.

2.2.5 Limitations in Research Visualization and Reporting

Visualization tools can make data exploration easier, but the quality of a report depends on the correctness of the underlying data and the suitability of the chosen visual representation. A dashboard may summarize results effectively while concealing important information about uncertainty, missing observations, measurement conditions, or sample selection.

General-purpose reporting tools also require a reliable connection to the underlying data source, suitable data models, correct calculations, and appropriate access controls. Power BI can provide an interactive reporting layer when integrated, but it does not independently replace the scientific calculations and validation procedures performed by the research system.

These limitations motivate a design in which data integrity, scientific processing, traceability, and visualization are treated as connected but distinct responsibilities. Table 2.1 summarizes the comparative features of existing approaches relative to the proposed GreenSynth Analytics platform.

2.3 Research Gap

The reviewed literature establishes that plant-mediated CuO synthesis is an active research area and that materials informatics platforms can support scientific data access, organization, and analysis. It also identifies continuing challenges related to reproducibility, heterogeneous data, metadata, and the interpretation of data-driven results.

Within the scope of the present project, the research gap is defined as the need to connect the specific experimental workflow for phytochemical-assisted CuO thin-film synthesis with a structured software environment for managing and analyzing the associated research data.

The identified project-level gaps are as follows:

1. **Experimental Traceability:** A need to maintain explicit relationships between synthesis parameters, experiment records, sample identities, characterization files, and derived properties.

Table 2.1: Comparison of Existing Materials Research and Informatics Approaches

Platform / Tool	**Primary Focus**	**Data Provenance**	**Automated Derivation (Tauc / Scherrer)**	**Lab Experimental Workflow**
Manual Tools (Excel / Origin)	Ad-hoc plotting and analysis	None (unversioned files)	Manual / subjective visual tangent line	Disconnected spreadsheets
The Materials Project	Computational materials database	System managed	Not applicable (computational DFT focus)	General computational repository
Citrination (Citrine)	Enterprise materials informatics	API dependent	Custom script dependent	Commercial cloud infrastructure
Materials Data Facility	Open research data sharing	Checksums on upload	Not applicable (storage and metadata focus)	General repository and discovery
GreenSynth Analytics (Proposed)	**Experimental lab + informatics**	**SHA-256 cryptographic hashes**	**Deterministic Python mathematical engines**	**Tailored thin-film synthesis and analysis**

2. **Structured Characterization Records:** A need to organize different types of characterization data within a common research workflow while preserving the original files and relevant metadata.

3. **Connection Between Measurements and Calculations:** A need to document how raw measurements are transformed into processed data and calculated properties, including the associated units and assumptions.

4. **Integrated Research Analysis:** A need to compare synthesis conditions and material properties through appropriate visualization and statistical analysis within the context of the project.

5. **Responsible Predictive Analysis:** A need to distinguish experimentally measured values from calculated, predicted, or synthetic values and evaluate predictive models using suitable validation procedures.

6. **Research-Specific Workflow Support:** A need to organize the scientific and software activities associated with the project within a platform that can be extended to additional materials or experimental workflows.

These gaps are the design motivations for GreenSynth Analytics. They should be interpreted as project-specific requirements derived from the literature and the needs of the current investigation, rather than proof that no existing software can perform similar tasks.

The contribution of the present work is therefore evaluated through the integration and implementation of the required research workflow, together with evidence of

software correctness and the scientific validity of any conclusions drawn from actual laboratory measurements.

2.4 Proposed System

The proposed system, **GreenSynth Analytics**, is a research data management and materials analysis platform developed in the context of phytochemical-assisted synthesis and characterization of CuO thin films. It aims to connect experimental records, characterization data, scientific calculations, and supported analytical workflows within a structured digital environment.

The system is designed to maintain relationships between projects, experiments, synthesis parameters, samples, characterization records, raw files, analysis operations, and derived material properties. This structure allows research information to be organized consistently and supports the traceability of analytical outputs to their corresponding experimental records.

The major functional areas of the proposed system are described below:

2.4.1 Research Project and Experiment Management

The platform organizes research projects and experimental records. Each experiment can be associated with its relevant synthesis conditions and samples, allowing results to be compared in the context of the conditions under which the material was prepared.

2.4.2 Sample and Characterization Management

The system links samples with their characterization records and associated raw files. Different characterization techniques may generate different types of data, so the platform must preserve relevant metadata and avoid treating distinct measurement types as interchangeable.

2.4.3 Scientific Data Processing

The platform supports scientific calculations and data-processing workflows where implemented. Processing procedures should identify the source data, applicable method, relevant parameters, units, and output so that derived values can be interpreted and checked.

2.4.4 Visualization and Comparative Analysis

GreenSynth Analytics provides a foundation for visualizing and comparing research data. Graphical analysis can help researchers explore distributions, relationships, and differences between experimental conditions. If Power BI is integrated, it can provide additional interactive reporting and dashboard functionality while the scientific calculations remain grounded in the underlying data-processing workflow.

2.4.5 Statistical Analysis and Machine Learning

Where implemented, statistical and machine-learning functionality can be used to investigate relationships between synthesis parameters and material properties. Model evaluation should include appropriate validation procedures, comparison with baseline approaches, and consideration of uncertainty and applicability to new conditions.

2.4.6 Experimental Design and Research Support

Design-of-experiments methods can support the systematic comparison of candidate experimental conditions. Proposed conditions or model-generated predictions are intended to guide further investigation rather than replace laboratory testing. Their usefulness must be assessed through suitable experimental validation.

2.4.7 Data Integrity and Reproducibility

The proposed system emphasizes the preservation of relationships between original records, raw files, processing operations, and derived outputs. It also requires appropriate access control and data validation to protect research records and maintain consistency.

Synthetic datasets may be used to test software functionality and analytical workflows. Such records must be identified clearly and must not be presented as experimental measurements.

Overall, GreenSynth Analytics is intended to connect materials research with structured data management and computational analysis. Its performance and contribution must be established through implementation evidence, software testing, and the scientific validity of results obtained from real laboratory measurements.

2.5 SDLC Model

The Software Development Life Cycle (SDLC) provides a structured approach to planning, designing, implementing, testing, deploying, and maintaining software systems. Selecting an appropriate development model is important because GreenSynth Analytics combines user-facing workflows, database operations, scientific processing, data visualization, and analytical functionality.

An **Iterative and Incremental Development Model** is suitable for a project of this nature because the platform can be developed in functional stages, with each stage reviewed and tested before further functionality is added. This approach allows requirements, data models, and analytical workflows to be refined as the experimental and software components become better understood.

2.5.1 Requirements Analysis

The first stage identifies the scientific workflow, user needs, functional requirements, non-functional requirements, and constraints. Requirements include the management of projects, experiments, samples, characterization data, analytical operations, and user access.

2.5.2 System and Database Design

The requirements are translated into the system architecture, database schema, API structure, user workflows, and access-control rules. The design establishes how experiments, samples, files, and calculated properties are connected.

2.5.3 Incremental Implementation

The platform is implemented in functional modules. Core data management and authentication can be established first, followed by experiment and sample workflows, characterization-file handling, scientific calculations, visualization, and supported analytical functions. Each increment should be checked against its requirements before additional functionality is introduced.

2.5.4 Testing and Verification

Each module is tested independently and in combination with other components. Testing includes input validation, database integrity, API behavior, access control, scientific calculations, file handling, and end-to-end research workflows.

2.5.5 Integration and Deployment

The modules are integrated into the complete platform and configured for the intended environment. Database connectivity, environment variables, storage access, user permissions, and operational behavior must be verified before the system is considered ready for use.

2.5.6 Evaluation and Maintenance

The system is reviewed against its requirements, and defects or usability issues are addressed. Future changes may include additional characterization workflows, analytical functions, report integrations, and support for additional materials.

2.5.7 Suitability of the Iterative and Incremental Model

The iterative and incremental model is appropriate when the project requires gradual integration of multiple technical components and repeated testing of research workflows. It supports early verification of core functionality and allows later analytical modules to be added without requiring every feature to be completed before testing begins.

However, iterative development does not remove the need for documented requirements, version control, testing, data integrity, and security review. These activities remain necessary throughout development. For GreenSynth Analytics, the development model provides a framework for organizing the software into manageable increments and validating each increment against the research requirements. The resulting platform must still be evaluated using documented tests and actual implementation evidence.

Chapter 3

REQUIREMENT SPECIFICATION

3.1 Functional Requirements

The functional requirements define the core operational capabilities provided by the GreenSynth Analytics platform to support experimental research workflows:

1. **User Authentication and Access Control:** Provide secure user registration, token-based session management, and role-based permissions (Administrator, Researcher, Viewer) to protect proprietary research data.
2. **Research Project Management:** Create, organize, and archive multi-investigator research projects, maintaining isolated project namespaces and team assignments.
3. **Experiment and Synthesis Logging:** Record comprehensive synthesis recipes, including chemical precursors (copper acetate monohydrate, ethanol, mulberry extract), operational parameters (substrate temperature, spray rate, carrier gas pressure), and real-time laboratory observations.
4. **Sample Identity and Lineage Tracking:** Assign persistent, unique identifiers to physical thin-film specimens and trace their hierarchical parentage back to specific synthesis experiments.
5. **Multi-Modal Characterization Ingestion:** Support multipart file uploads for raw instrumental data across five distinct techniques: X-ray diffraction (XRD), ultraviolet-visible (UV-Vis) spectroscopy, Fourier-transform infrared (FTIR) spectroscopy, scanning electron microscopy (SEM), and current-voltage (I–V) transport sweeps.
6. **Automated Scientific Derivation Engines:** Execute deterministic mathematical processing pipelines on uploaded raw spectra to calculate physical material properties:
   - Automatic polynomial background subtraction and Scherrer crystallite size (D) calculation from XRD peaks.
   - Automated photon energy ($h\nu$) conversion, Tauc coordinate transformation ($(\alpha h\nu)^2$), and band gap (E_g) extraction via maximum first-derivative tangent fitting.
   - Linear least-squares regression over I–V sweeps to calculate electrical resistance (R), sheet resistance (R_s), and conductivity (σ).
7. **Interactive Data Visualization:** Render interactive spectral curves, multi-sample overlays, parameter distribution histograms, and property correlation scatter plots directly within the user interface.

8. **Predictive Machine-Learning Benchmarking:** Train, evaluate, and benchmark cross-validated regression models predicting functional properties from synthesis variables, guarded by automated target data leakage prevention mechanisms.

9. **Standardized Scientific Dossier Generation:** Compile automated, publication-ready research reports in multi-page PDF format utilizing ReportLab, capturing data provenance, mathematical derivations, and tabular summaries.

10. **Cryptographic Data Integrity Verification:** Generate SHA-256 cryptographic digests upon file upload to ensure tamper detection and guarantee research reproducibility.

3.2 Non Functional Requirements

The platform adheres to stringent software quality standards to ensure usability, security, and scientific reliability:

- **Security and Access Protection:** All API communications must be encrypted over HTTPS. User authentication tokens must utilize industry-standard cryptographic signatures (JWT), and raw data uploads must be fingerprinted using SHA-256 digests to detect unauthorized post-measurement alterations.

- **System Reliability and Fault Tolerance:** The backend services must gracefully handle malformed data files, irregular instrumental headers, and network interruptions without crashing, returning informative HTTP error responses with actionable diagnostics.

- **Performance and Response Latency:** Standard database queries and API endpoints must respond within 200 ms under typical laboratory loads. Deterministic scientific derivation routines (Scherrer, Tauc, Ohmic regression) must execute in under 50 ms per spectral dataset.

- **Usability and Interface Clarity:** The user interface must provide responsive, intuitive navigation across desktop and tablet devices, displaying progress indicators during file uploads and presenting scientific charts with accessible units and legends.

- **Maintainability and Modularity:** The software architecture must follow a modular, service-oriented design with clear separation between API routing, business logic, data models, and derivation engines, supported by comprehensive automated test suites.

- **Scalability:** The relational schema must support thousands of experiments, samples, and high-frequency spectral arrays without degradation in query performance, leveraging database indexing and connection pooling.

- **Data Integrity and Traceability:** Relational integrity must be enforced via foreign keys, unique constraints, and database cascading rules. Every derived property must maintain an unbroken audit link to its parent sample, raw file, and mathematical algorithm version.

- **Reproducibility:** Scientific derivations must be purely deterministic: processing the same raw data file with identical parameters must yield identical calculated numerical outputs regardless of execution environment.

3.3 Hardware Requirements

Because this project encompasses both laboratory-based materials synthesis and computational software engineering, hardware requirements are specified across computational resources and wet-lab instrumentation.

3.3.1 Computing and Server Hardware Specifications

The computational infrastructure required for hosting, executing, and interacting with GreenSynth Analytics is detailed in Table 3.1.

Table 3.1: Computing and Server Hardware Requirements

Component	**Suggested Requirement**
Processor	Modern multi-core x86-64 processor (Intel Core i5 / AMD Ryzen 5 or higher)
RAM	Minimum 8 GB; 16 GB recommended for multi-model ML benchmarking
Storage	Minimum 10 GB available SSD storage for database and raw characterization files
Display	1366×768 resolution or higher (1920×1080 recommended for dashboards)
Network	Broadband internet / LAN connection for remote database and API access

3.3.2 Laboratory Synthesis and Characterization Instrumentation

The experimental investigation requires precision wet-laboratory apparatus and analytical instruments for phytochemical extraction, spray pyrolysis deposition, and thin-film characterization, as summarized in Table 3.2.

3.4 Software Requirements

The software architecture leverages modern, open-source programming frameworks, database engines, and scientific computing packages, detailed in Table 3.3.

3.5 System Environment

GreenSynth Analytics operates as a multi-tier, web-based client-server system designed to bridge laboratory measurement stations with centralized computing and analytical reporting:

Table 3.2: Laboratory Experimental Apparatus and Analytical Instrumentation

Apparatus / Instrument	**Model / Specifications**	**Experimental Function in Project**
Soxhlet Extraction Assembly	Borosilicate glass, heating mantle (80°C)	Ethanolic extraction of active phytochemicals from mulberry leaves
Spray Pyrolysis Deposition Unit	Pneumatic aerosol nozzle, custom heating stage (200–450°C)	Atomization and pyrolytic deposition of CuO thin films on glass
Digital pH Meter	Bionexis BNPH002 (±0.01 pH precision)	pH monitoring during precursor solution preparation and titration
Digital Conductivity Meter	LJ-alpha 06	Electrical conductivity measurement of phytochemical and precursor solutions
UV-Vis Spectrophotometer	Digital Spectrophotometer (190–1100 nm)	Optical transmittance and absorption spectroscopy for band gap calculation
UV Fluorescence Cabinet	Dual wavelength (254 nm/365 nm)	Visual fluorescence examination of extracted phytochemical fractions
Digital Melting Point Unit	Precision capillary heating block	Thermal characterization and melting point verification
Centrifuge & Drying Oven	Benchtop centrifuge; thermostated hot air oven	Precipitate separation, glass substrate cleaning, and sample drying

- **Presentation Tier (Frontend Client):** Built using React 18 with TypeScript, delivering an interactive Single Page Application (SPA). The frontend communicates with the backend exclusively via asynchronous RESTful HTTP endpoints, handling user authentication, tabular experiment views, interactive charts, and file upload forms.

- **Application and Logic Tier (Backend Service):** Powered by Python 3 and FastAPI, orchestrating business logic, security middleware, parameter validation via Pydantic schemas, and dispatching mathematical tasks to NumPy and SciPy. The backend also hosts the Scikit-learn regression benchmarking pipeline and the ReportLab PDF generation service.

- **Data Tier (Relational Database & File Vault):** Structured records (projects, experiments, samples, parameters, calculated properties) are maintained in PostgreSQL or SQLite using a normalized Third Normal Form (3NF) relational schema managed by SQLAlchemy 2.0 ORM. Raw instrumental ASCII, CSV, and image files are stored in an organized local file vault, referenced by immutable path keys and cryptographically verified against their SHA-256 hashes.

- **External Analytics Tier (Reporting & Dashboards):** Optionally integrated with Microsoft Power BI via direct database connectors or CSV data

Table 3.3: Software Technology Stack and Environment Requirements

Software / Technology	**Role and Purpose in GreenSynth Platform**
Operating System	Windows 10/11, Ubuntu Linux 22.04 LTS, or macOS for development and hosting
Python 3.10+	Core runtime for backend web services, data processing, and scientific derivations
FastAPI	High-performance asynchronous RESTful API framework with automatic OpenAPI docs
React 18 & TypeScript	Component-based interactive frontend for experiment tracking and visual dashboards
PostgreSQL / SQLite	Relational data persistence with 3NF schema for projects, experiments, and samples
SQLAlchemy 2.0 & Pydantic	Object-Relational Mapping (ORM) and strict runtime data validation schemas
NumPy, Pandas, SciPy	Deterministic scientific calculation engines (Scherrer, Tauc, Ohmic linear regression)
Scikit-learn	Machine-learning pipelines with cross-validation and target leakage guards
ReportLab	Programmatic PDF engine generating audit-ready, standardized scientific dossiers
Plotly & Matplotlib	Dynamic interactive web charts and publication-grade static spectral visualizations
Git	Distributed version control and source code management
Microsoft Power BI	Optional enterprise business intelligence and multi-filter analytical reporting

feeds, providing executive summary dashboards, multi-dimensional filtering, and high-level cross-project comparisons while keeping the scientific derivation logic strictly within the Python core.

The system supports seamless transition between isolated local development (SQLite, Vite dev server, localhost FastAPI) and networked production environments (PostgreSQL, Uvicorn ASGI behind Nginx reverse proxy, static React distribution) via standardized environment variables and configuration files.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

The system architecture of GreenSynth Analytics is engineered around a modular, decoupled four-tier paradigm designed specifically to address the demanding computational, provenance, and data-integrity requirements of modern materials informatics. Scientific research environments require deterministic reproducibility, immutable provenance tracking, and seamless separation between compute-intensive numerical modeling and high-throughput interactive user interfaces.

The four architectural tiers comprise:

1. **Presentation Tier (Client Layer):** Built with React 18 and TypeScript, this layer delivers a responsive single-page application (SPA). It provides interactive spectral visualization dashboards (leveraging Plotly.js for zoomable, pannable XRD and UV-Vis diffractograms/absorbance spectra), dynamic experiment logging forms, and predictive confidence cards. Communication with the backend occurs asynchronously through secure RESTful endpoints using Axios over HTTPS.

2. **Application and API Tier (Service Layer):** Powered by FastAPI (Python 3.11) running asynchronously on Uvicorn ASGI workers. This tier handles HTTP request routing, JWT-based authentication, RBAC authorization, Pydantic schema validation, and transaction orchestration. It provides deterministic rate-limiting and structured JSON error responses.

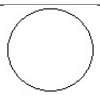
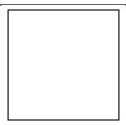
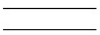
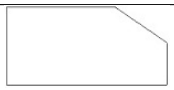
3. **Computation and Scientific Engine Tier (Domain Logic):** Implements vectorized scientific algorithms using NumPy and SciPy. This tier executes baseline subtraction, Savitzky-Golay filtering, XRD crystallite size computation via the Debye-Scherrer formulation, optical band gap extraction via Tauc plot linear extrapolation, and electrical resistivity/conductivity derivation from linear I-V regressions. It also encapsulates the scikit-learn machine learning pipeline and the orthogonal array Design of Experiments (DOE) optimization generator.

4. **Data Persistence Tier (Storage Layer):** Divided into a relational database layer (PostgreSQL 15 / SQLite) managed via SQLAlchemy 2.0 ORM with Alembic schema migrations, and an immutable local file vault. Raw spectral text/CSV files uploaded from lab instruments are hashed with cryptographic SHA-256 digests prior to storage on disk, guaranteeing tamper-evident audit trails.

4.2 Analysis model

The Analysis Model formally structures the data transformations and functional boundaries of the system using Data Flow Diagrams (DFDs) and Entity-Relationship modeling. The standard graphical notations utilized across all DFD levels are summarized in Table 4.1.

Table 4.1: Data Flow Symbols

Sr. No.	Symbol	Description
1		A **function** (process bubble) represents algorithmic or operational data transformation applied to incoming data flows.
2		An **external entity** represents an external source or destination (human actor or hardware instrument) outside system boundaries.
3		A **data flow** (directed arrow) indicates the movement and direction of information between processes, data stores, and entities.
4		A **data store** (parallel lines) represents a persistent physical or logical data repository (relational table or disk vault).
5		An **output interface** represents human-computer interaction displays, reports, and analytical presentation targets.

4.2.1 ER Diagram

The relational data model of GreenSynth Analytics is normalized to Third Normal Form (3NF) to eliminate update anomalies, ensure referential integrity, and maintain strict lineage between physical laboratory samples and digital analytical runs.

Figure 5.1 illustrates the complete Entity-Relationship diagram in Crow's Foot notation, detailing primary keys (PK), foreign keys (FK), and inter-table cardinalities.

The database schema is organized across five cohesive architectural domains:

1. **Access and Governance Domain:** The `USERS` entity tracks system operators, passwords (bcrypt hashes), and assigned roles. Each user belongs to one or more `RESEARCH_GROUPS` via the associative table `GROUP_MEMBERSHIPS`. All administrative and state-changing actions generate timestamped entries in `AUDIT_LOGS`.

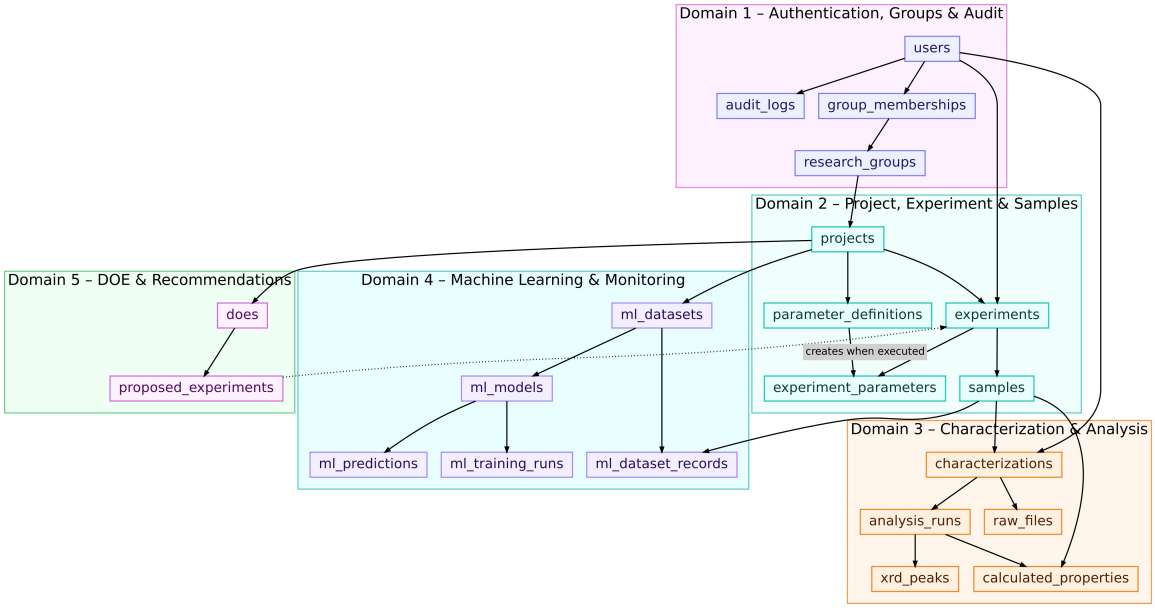


Figure 4.1: GreenSynth Analytics High-Level Domain Conceptual Architecture

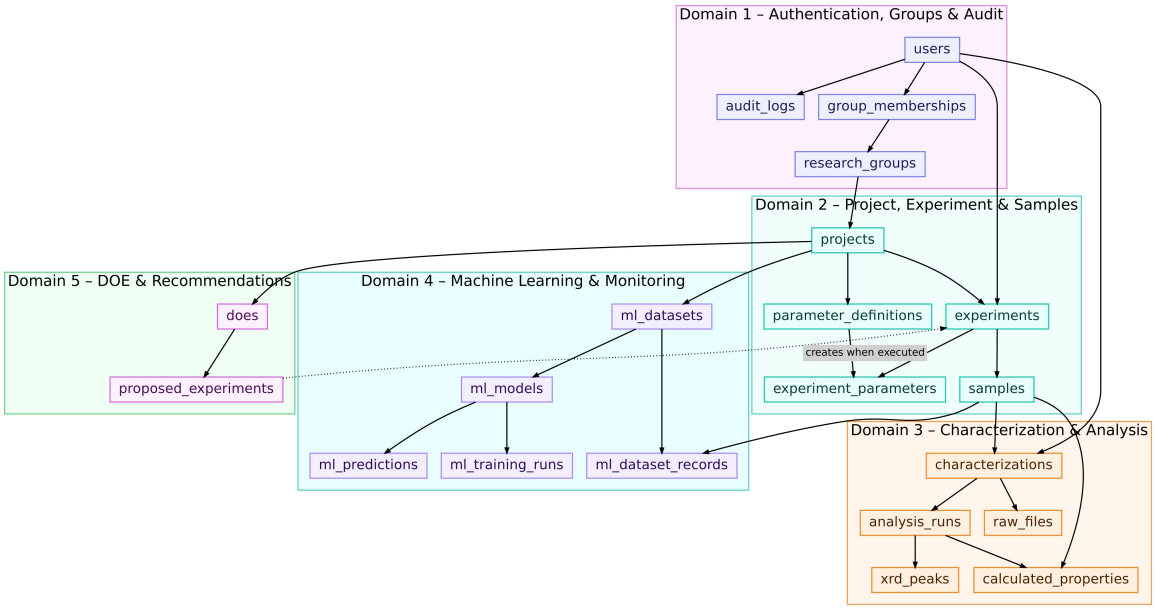


Figure 4.2: Detailed Relational Entity-Relationship Diagram of GreenSynth Analytics

2. **Project, Experiment, and Sample Hierarchy:** A `PROJECTS` entity establishes the overarching research scope (e.g., green-synthesized thin-film semiconductors). A project owns dynamic `PARAMETER_DEFINITIONS` (which configure admissible minimum and maximum bounds for physical factors) and multiple `EXPERIMENTS`. Each experiment records specific synthesis run conditions across up to 15 factors in `EXPERIMENT_PARAMETERS` and produces one or more physical `SAMPLES`.

3. **Characterization and Analysis Engine Domain:** Each sample is measured across multiple characterization techniques stored in `CHARACTERIZATIONS`. The raw data streams from analytical instruments are preserved in `RAW_FILES`, which maintains the original filename, storage URI, and a SHA-256 cryptographic digest. Computational analysis runs (`ANALYSIS_RUNS`) execute deterministic derivation routines, generating extracted `XRD_PEAKS` and physical parameters stored in `CALCULATED_PROPERTIES` (e.g., crystallite size D, optical band gap E_g, electrical resistivity ρ, and conductivity σ).

4. **Machine Learning and Model Registry Domain:** Curated synthesis and characterization vectors are compiled into `ML_DATASETS` and individual rows in `ML_DATASET_RECORDS`. Supervised regressors are tracked in `ML_MODELS`, their 5-fold cross-validation scores documented in `ML_TRAINING_RUNS`, and predictive inferences logged in `ML_PREDICTIONS`.

5. **Design of Experiments (DOE) Domain:** Systematic experimental optimization matrices are configured in `DOES`. When an orthogonal array or response surface design is computed, proposed synthesis conditions are written to `PROPOSED_EXPERIMENTS`, which can subsequently be instantiated as active lab experiments.

4.2.2 DFD Level 0

The Level 0 Data Flow Diagram (Context-Level Diagram), depicted in Figure 4.3, models the boundaries of GreenSynth Analytics relative to external entities.

The system interacts with four primary external entities:

1. **Materials Researcher / Scientist:** Supplies project definitions, chemical precursors, spray deposition parameters, DOE optimization targets, and ML query inputs. In return, the researcher receives derived physical properties (D, E_g, σ, ρ), interactive spectral curves (XRD diffractograms, Tauc plots, I-V traces), model evaluation leaderboards, and exported PDF/CSV dossiers.

2. **Lab Technician / Instrument Operator:** Provides physical sample metadata, uploads raw instrument characterization files (.csv, .txt), and configures manual baseline correction overrides. The system returns SHA-256 upload receipts, file integrity verification confirmations, and automated parsing status flags.

3. **Analytical Lab Instruments (XRD, UV-Vis, FTIR, Four-Probe):** Serve as physical data sources emitting raw spectral intensity curves, optical absorbance arrays, and electrical current-voltage sweeps. The platform issues automated file reception receipts and checksum validations.

4. **System Administrator / Lab Director:** Configures user credentials, role-based access control policies, and research group privileges. Receives system health diagnostics, performance monitoring logs, and comprehensive tamper-evident audit trails.

4.2.3 DFD Level 1

The Level 1 Data Flow Diagram, shown in Figure 4.4, decomposes Process 0.0 into five interconnected functional sub-processes and maps their interactions with six persistent data stores (**D1** to **D6**).

The five core Level 1 processes operate as follows:

1. **Process 1.0 (Authentication & Access Control):** Validates incoming researcher and administrator credentials against Data Store **D1 (users & groups)**. Upon verification, it issues encrypted JWT access tokens that enforce role-based permissions across all API endpoints.

2. **Process 2.0 (Experiment Setup & Parameter Management):** Facilitates project configuration, spray pyrolysis deposition variable entry (15 distinct synthesis parameters), and sample cataloging. It writes experimental metadata to Data Store **D2 (experiments & samples)** and numeric factor values to Data Store **D3 (experiment_parameters)**.

3. **Process 3.0 (Characterization & Scientific Engine):** Ingests raw spectral files, calculates SHA-256 checksums, and executes deterministic scientific derivations (Debye-Scherrer crystallite sizing, Tauc optical band gap extrapolation, and electrical resistivity/conductivity derivation). Parsed files and computed properties are persisted in Data Store **D4 (raw_files & calculated_properties)**.

4. **Process 4.0 (Machine Learning & Prediction Pipeline):** Extracts feature vectors from **D3** and ground-truth targets from **D4**. It trains 5-fold cross-validated regressors, verifies applicability domain boundaries, and saves trained models, hyperparameter configurations, and inference outputs to Data Store **D5 (ml_models & predictions)**.

5. **Process 5.0 (DOE Optimization & Reporting Service):** Uses experimental results and ML surrogate models to construct orthogonal experimental design matrices (L_9/L_{16}) aimed at property optimization (e.g., maximizing electrical conductivity σ). The generated candidate matrices are written to Data Store **D6 (does & proposed_experiments)** and rendered as publication-ready dossiers.

4.2.4 DFD Level 2

Level 2 Data Flow Diagrams expand the core computational subsystems into detailed algorithmic sub-processes. GreenSynth Analytics defines two primary Level 2 subsystems: the Characterization & Derivation Subsystem (Process 3.0) and the Machine Learning & DOE Optimization Subsystem (Process 4.0).

DFD Level 2: Subsystem 3.0 (Characterization & Scientific Engine)

Figure 4.5 details the data flow within Process 3.0 as raw experimental files pass from initial ingestion to derived physical constants and curated ML feature matrices.

The sub-processes in Subsystem 3.0 include:

1. **Process 3.1 (File Ingestion & SHA-256 Checksum):** Validates file format (.csv, .txt), computes a SHA-256 hash to ensure provenance, stores the binary stream into the secure file vault, and registers metadata in Data Store **D1 (raw_files)**.

2. **Process 3.2 (XRD Peak Detection & Scherrer Engine):** Parses 2θ-intensity arrays, removes continuous background radiation, identifies diffraction peaks, fits Gaussian/Lorentzian profiles to compute Full Width at Half Maximum (β), and calculates crystallite size (D) via the Debye-Scherrer equation ($D = \frac{K\lambda}{\beta \cos\theta}$). Peak locations are written to **D2 (xrd_peaks)** and crystallite size to **D3 (calculated_properties)**.

3. **Process 3.3 (UV-Vis Tauc Band Gap Extrapolation Engine):** Transforms wavelength (λ) and absorbance (A) into photon energy ($h\nu$) and absorption coefficient (α). Constructs $(\alpha h\nu)^2$ vs. $h\nu$ arrays, locates the maximum linear slope region, and extrapolates the tangent line to the abscissa intercept ($\alpha h\nu = 0$) to compute direct optical band gap (E_g), persisted in **D3**.

4. **Process 3.4 (Electrical I-V Ohmic Conductivity Engine):** Accepts current-voltage sweeps, executes linear least-squares regression to extract sheet resistance (R), and applies sample geometry (film thickness t and contact dimensions) to calculate electrical resistivity (ρ) and electrical conductivity ($\sigma = 1/\rho$), stored in **D3**.

5. **Process 3.5 (ML Feature Assembler & Leakage Guard):** Joins physical synthesis parameters from **D4 (experiment_parameters)** with verified target properties from **D3**, enforces a strict leakage prevention rule (excluding intermediate characterizations from input features), and packages clean tensors into Data Store **D5 (ml_datasets & records)**.

DFD Level 2: Subsystem 4.0 (Machine Learning & DOE Optimization)

Figure 4.6 illustrates the data flows governing feature normalization, cross-validation, champion model selection, applicability domain validation, and orthogonal experimental design generation.

The sub-processes in Subsystem 4.0 execute in the following sequence:

1. **Process 4.1 (Feature Preprocessing & Leakage Guard):** Pulls target properties (σ) from **D3** and synthesis variables from **D4**. Performs MinMax feature scaling and standardization, guaranteeing that no target-correlated characterization variables leak into the input space. Persists the normalized matrix in **D5 (ml_datasets)**.

2. **Process 4.2 (5-Fold Cross-Validation & Model Training):** Partitions the training matrix into 5 stratified folds. Fits Random Forest Regressor, Gradient Boosting Regressor, and XGBoost models. Model binaries are serialized

(.joblib) into local storage, and fold metrics (R^2, RMSE, MAE) are recorded in **D6 (ml_models & runs)**.

3. **Process 4.3 (Model Evaluation & Champion Selection):** Compares out-of-fold generalization metrics across candidate architectures. Selects the champion regressor based on lowest cross-validated RMSE and highest R^2, updating the model registry status flag.

4. **Process 4.4 (Prediction Inference & Domain Bounds Check):** Accepts arbitrary synthesis queries from researchers, projects candidate vectors into the trained feature space, checks Mahalanobis distance / bounding-box limits to verify applicability domain validity, and calculates point predictions with 95% confidence intervals, saving results to **D7 (ml_predictions)**.

5. **Process 4.5 (DOE Matrix & Candidate Recommendation):** Ingests optimization goals (e.g., maximize conductivity σ), explores high-scoring predicted regimes, and constructs an orthogonal experimental array (L_9/L_{16}) that balances parameter space exploration with exploitation. Writes candidate runs into Data Store **D8 (does & proposed_experiments)**.

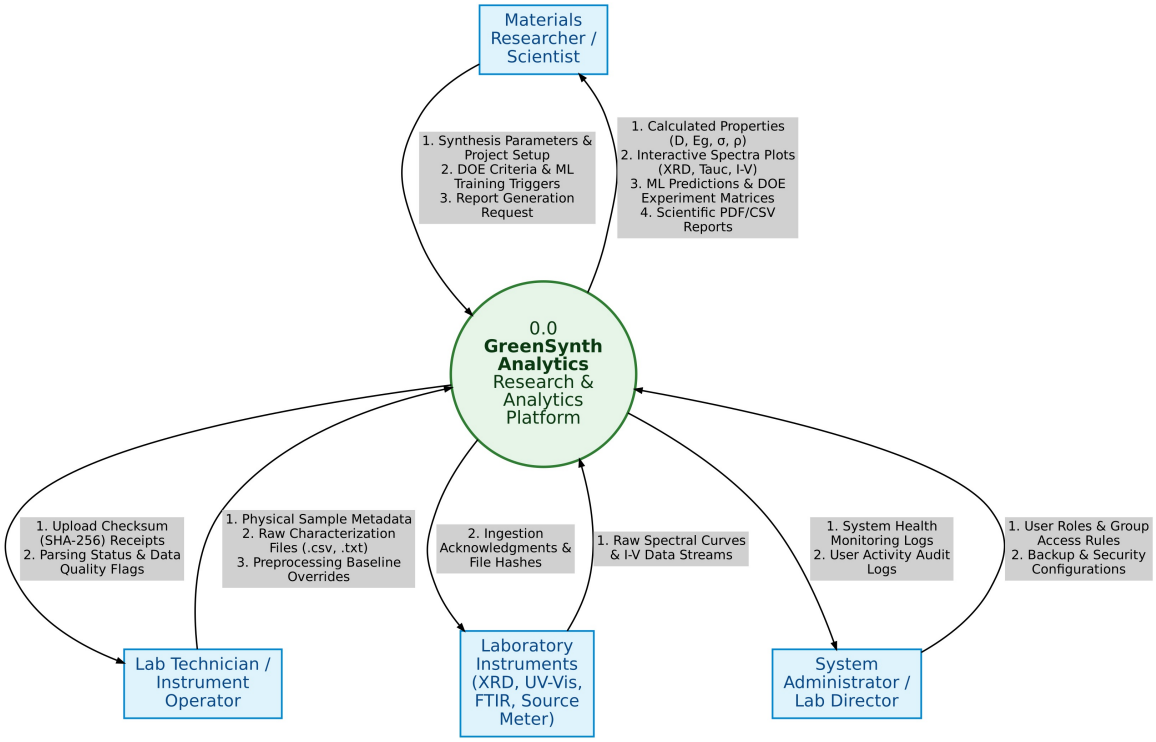


Figure 4.3: DFD Level 0 (Context Level Diagram) of GreenSynth Analytics

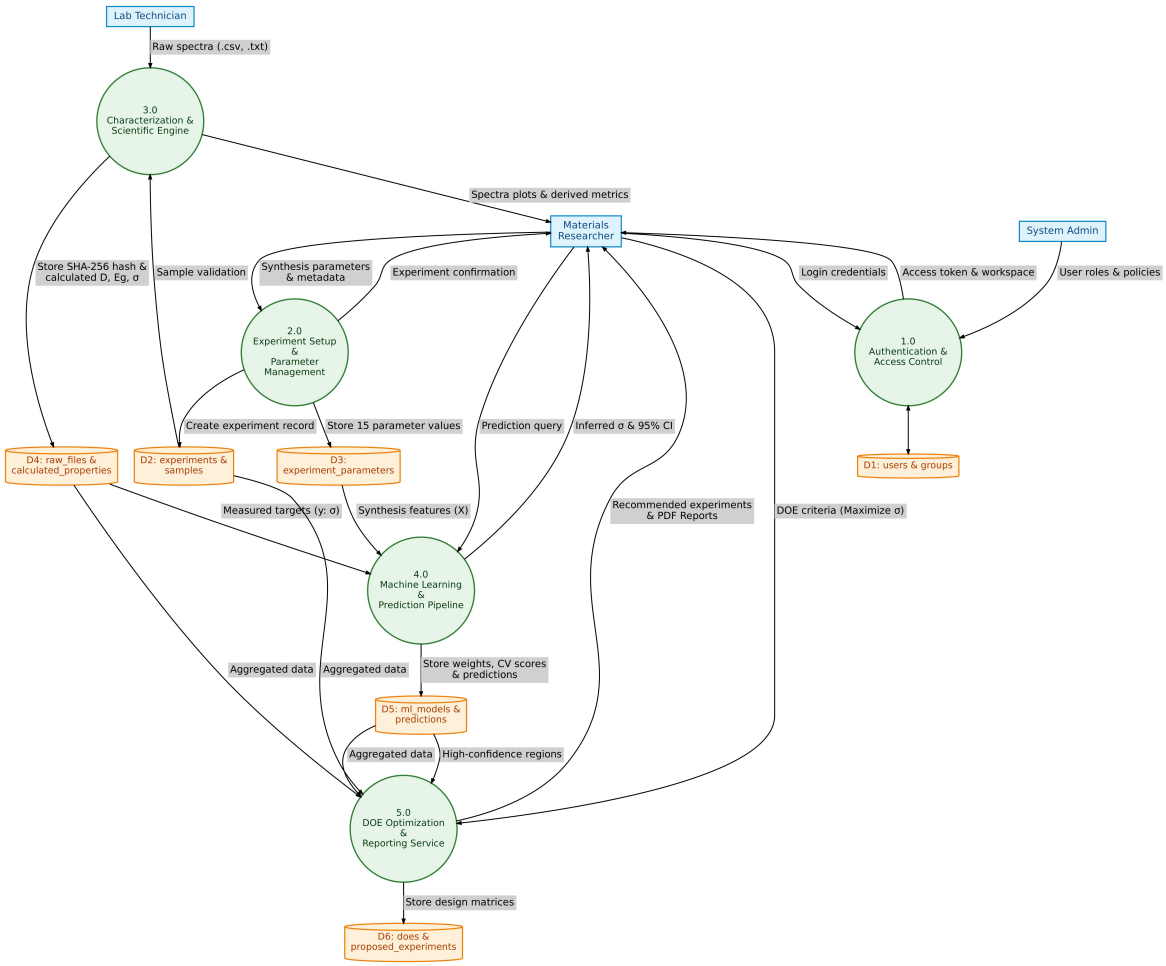


Figure 4.4: DFD Level 1 Diagram of GreenSynth Analytics

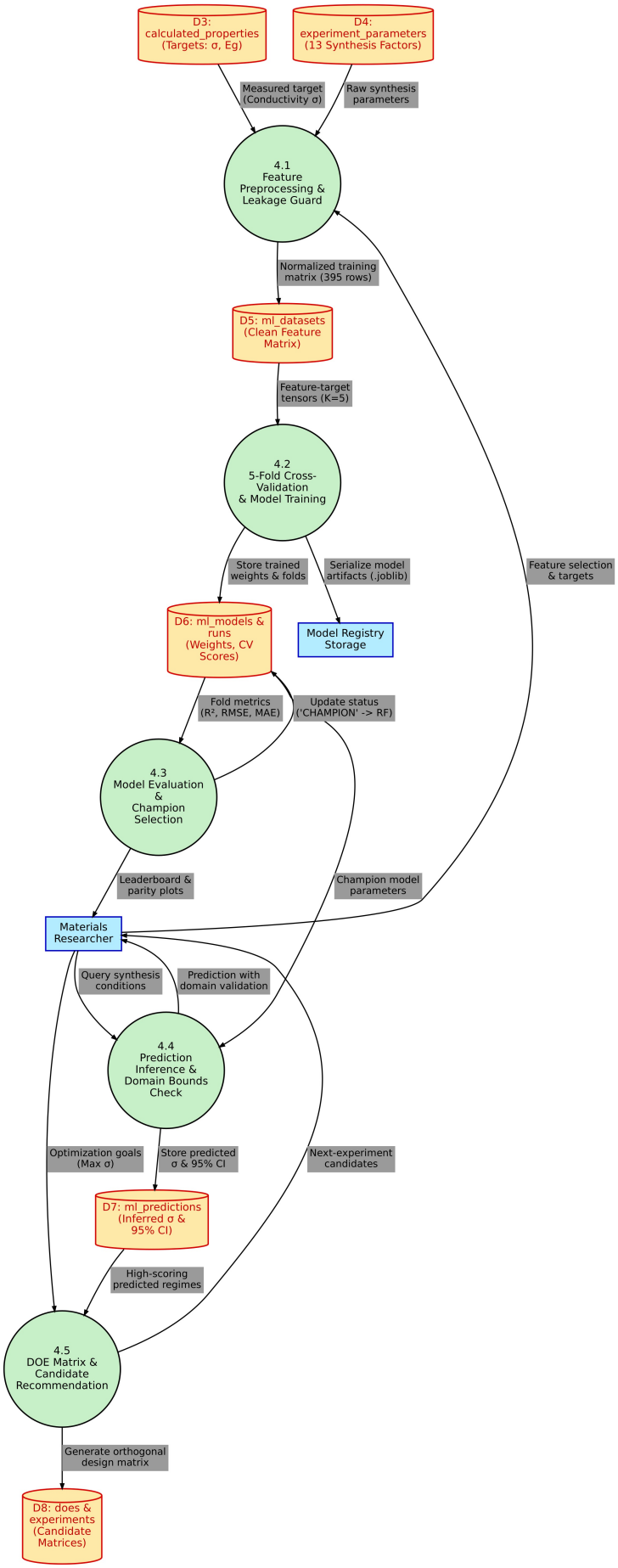


Figure 4.5: DFD Level 2: Subsystem 3.0 (Characterization & Scientific Derivation Engine)

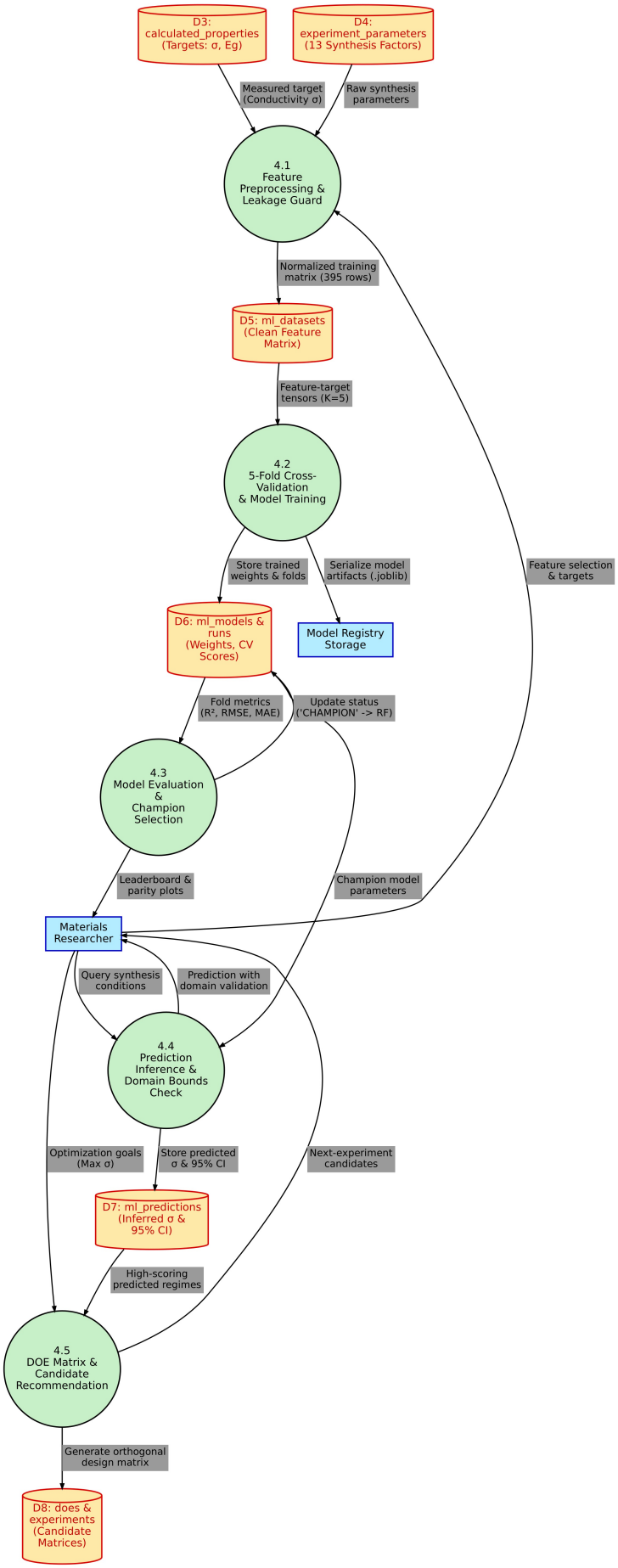


Figure 4.6: DFD Level 2: Subsystem 4.0 (Machine Learning Pipeline & Design of Experiments)

4.3 UML Diagrams

Unified Modeling Language (UML) diagrams provide an object-oriented structural and behavioral specification of GreenSynth Analytics.

4.3.1 Use Case Diagram

The Use Case Diagram shown in Figure 4.7 models system functionality from the perspective of external actors, delineating system boundaries and dependency stereotypes ($\ll$ *include* $\gg$ and $\ll$ *extend* $\gg$).

The system defines four actors and twenty use cases organized across five functional packages:

1. **Materials Researcher:** Interacts with **UC1 (Login & Authentication)**, **UC2 (Configure Project)**, **UC3 (Define 15 Spray Parameters)**, **UC4 (Register Experiment Runs)**, **UC11 (Visualize Spectra Plots)**, **UC12 (Assemble ML Training Dataset)**, **UC14 (Train 5-Fold CV Regressors)**, **UC15 (Predict Conductivity)**, **UC17 (Generate DOE Design Matrix)**, and **UC18 (Export PDF/CSV Reports)**.
2. **Lab Technician:** Interacts with **UC1**, **UC5 (Catalog Physical Samples)**, and **UC6 (Upload Raw Spectra Files)**.
3. **Scientific Engine (Automated Background Worker):** Executes **UC8 (Calculate Scherrer Crystallite Size)**, **UC9 (Extrapolate Tauc Band Gap)**, and **UC10 (Derive Ohmic Conductivity)**.
4. **System Administrator:** Manages **UC19 (Manage Roles & Groups)** and **UC20 (Inspect Audit Trails)**.
5. **Key Stereotype Dependencies:**
   - **UC6** $\ll$ *include* $\gg$ **UC7 (Verify SHA-256 Checksum)**: File storage cannot proceed without generating a cryptographic integrity digest.
   - **UC12** $\ll$ *include* $\gg$ **UC13 (Guard Against Target Leakage)**: Compiling an ML dataset mandates automated screening to exclude target-derived characterization metrics.
   - **UC15** $\ll$ *extend* $\gg$ **UC16 (Check Applicability Domain)**: Predictive conductivity inference dynamically triggers domain boundary verification when candidate features approach the bounds of the training data.

4.3.2 Class Diagram

The UML Class Diagram in Figure 4.8 depicts the static object structure of GreenSynth Analytics, illustrating class attributes, method signatures, access visibility, and structural relationships (associations, aggregations, compositions, and dependencies).

The core domain classes include:

- `Project`: Top-level container representing an experimental campaign. Possesses composition relationships with `ParameterDefinition` and `Experiment`, and aggregations with `MLDataset` and `DOEService`.

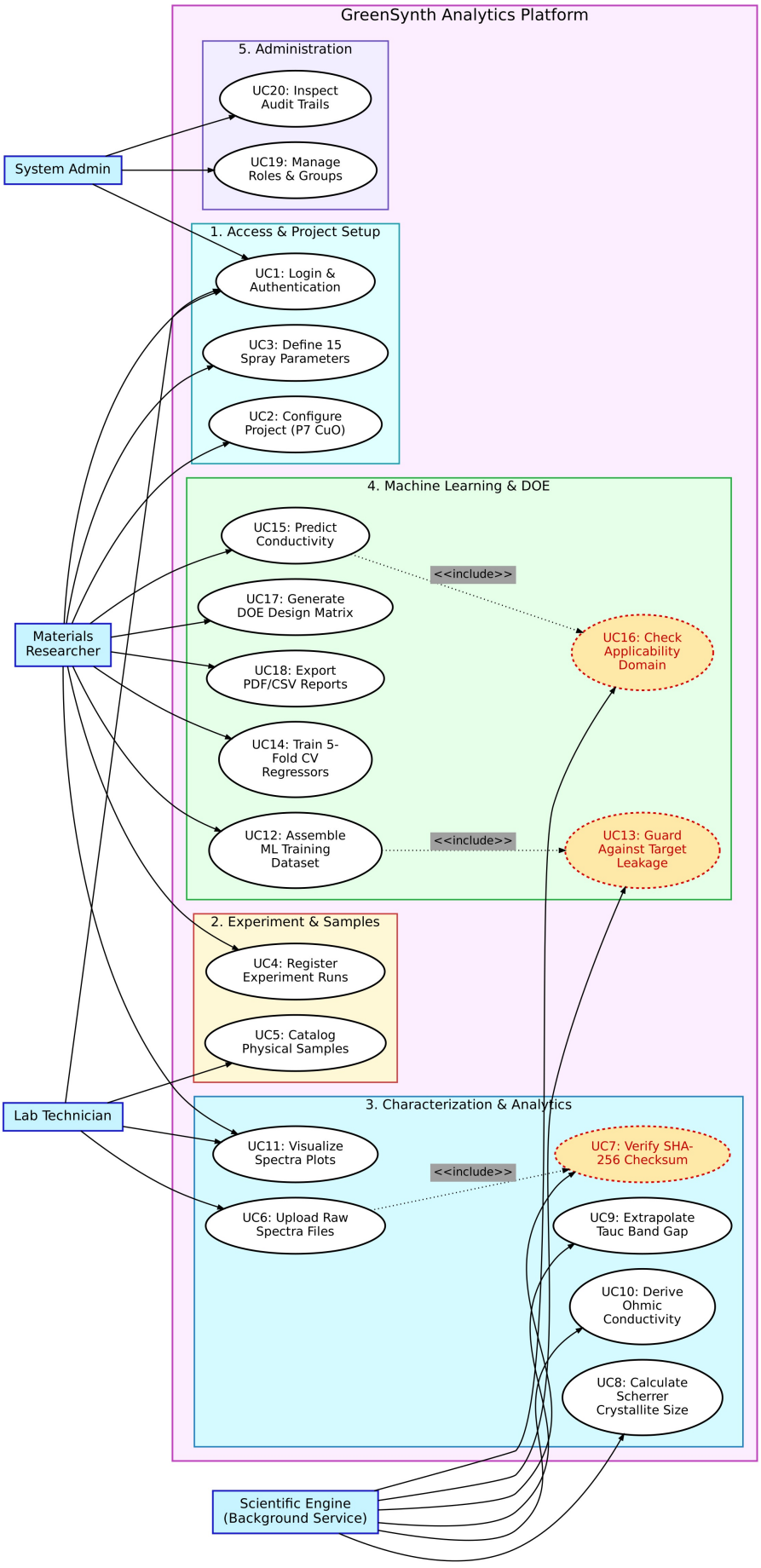


Figure 4.7: UML Use Case Diagram of GreenSynth Analytics

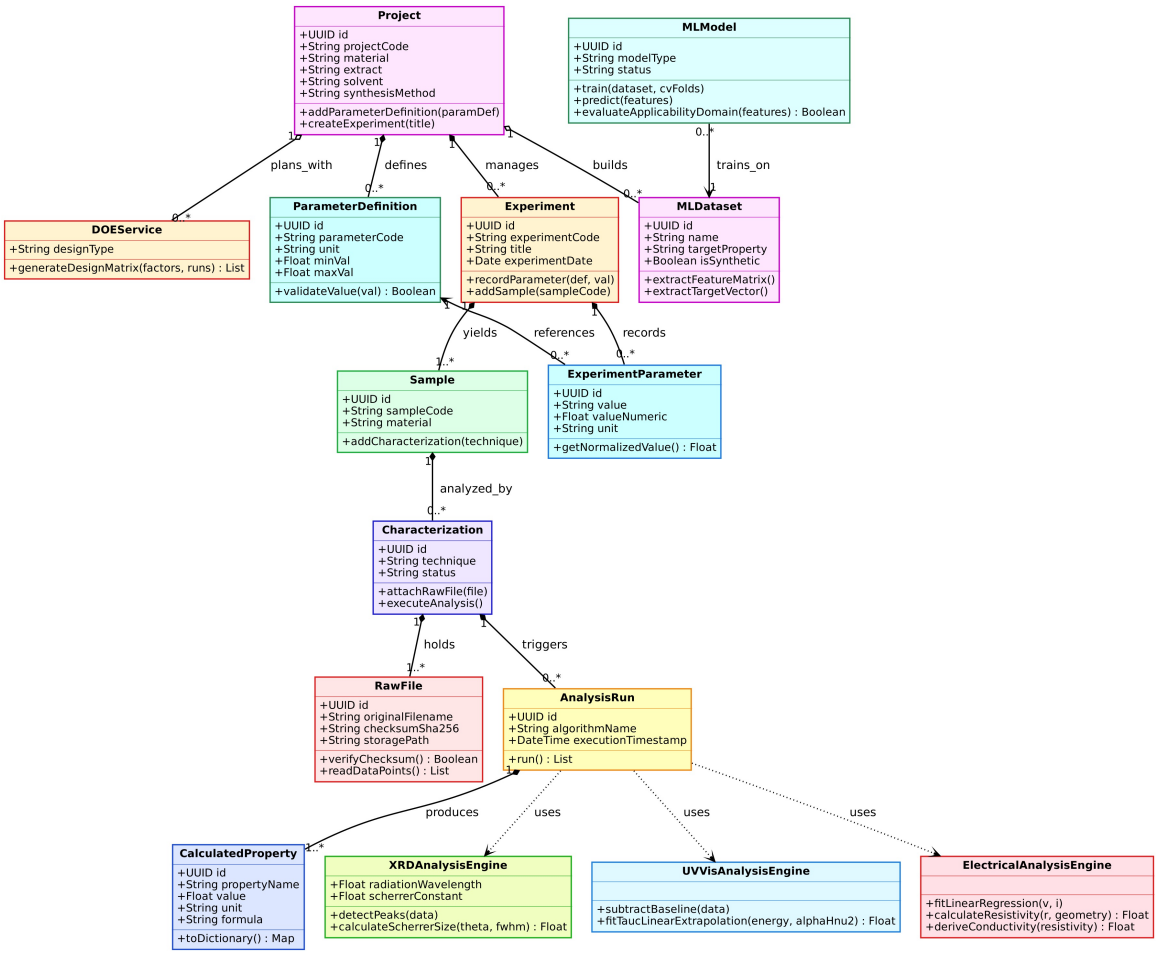


Figure 4.8: UML Class Diagram of GreenSynth Analytics

- `ParameterDefinition` & `ExperimentParameter`: Maintain dynamic synthesis parameter definitions (names, units, bounds) and their associated numerical values for individual experiment runs.
- `Experiment` & `Sample`: Model physical synthesis operations and the resulting thin-film specimens. An experiment yields one or more `Sample` instances.
- `Characterization`, `RawFile`, and `AnalysisRun`: Represent the analytical measurement hierarchy. A characterization holds uploaded raw files (verifying their SHA-256 hash) and initiates `AnalysisRun` executions.
- **Specialized Scientific Engines:** `AnalysisRun` delegates computation to specialized statutory engines: `XRDAnalysisEngine` (peak identification and Debye-Scherrer sizing), `UVVisAnalysisEngine` (baseline subtraction and Tauc tangent extrapolation), and `ElectricalAnalysisEngine` (linear regression and resistivity/conductivity calculation). Results are instantiated as `CalculatedProperty` records.
- `MLDataset`, `MLModel`, and `DOEService`: Encapsulate predictive intelligence, 5-fold cross-validation training, domain checks, and orthogonal array candidate experiment generation.

4.3.3 Sequence Diagram

The UML Sequence Diagram depicted in Figure 4.9 details the dynamic runtime interactions among the researcher, frontend interface, REST API, storage engine, scientific engine, relational database, and ML inference service.

The interaction executes through four sequential phases:

1. **Phase 1: File Upload and Cryptographic Validation (Steps 1–8):**
   - The Researcher selects a physical sample and uploads a raw characterization file (e.g., `P7_XRD_CuO.csv`) via the React Frontend.
   - The Frontend dispatches an asynchronous HTTP POST request to `/api/v1/characterizations/upload`.
   - The FastAPI Characterization API streams the file to the File Storage Service, computes its SHA-256 digest, and persists the raw file in the storage vault.
   - File metadata and cryptographic hash digests are committed to `raw_files` and `characterizations` in the database.
2. **Phase 2: Automated Scientific Derivation (Steps 9–14):**
   - The API invokes `execute_analysis(technique="XRD", file_path)` on the Scientific Engine.
   - The engine executes baseline subtraction, isolates the major diffraction peak (e.g., at $2\theta = 38.74^\circ$), and applies the Debye-Scherrer formula to determine crystallite size ($D = 19.10$ nm).
   - The resulting peak parameters and derived crystallite size are committed to `analysis_runs` and `calculated_properties`.

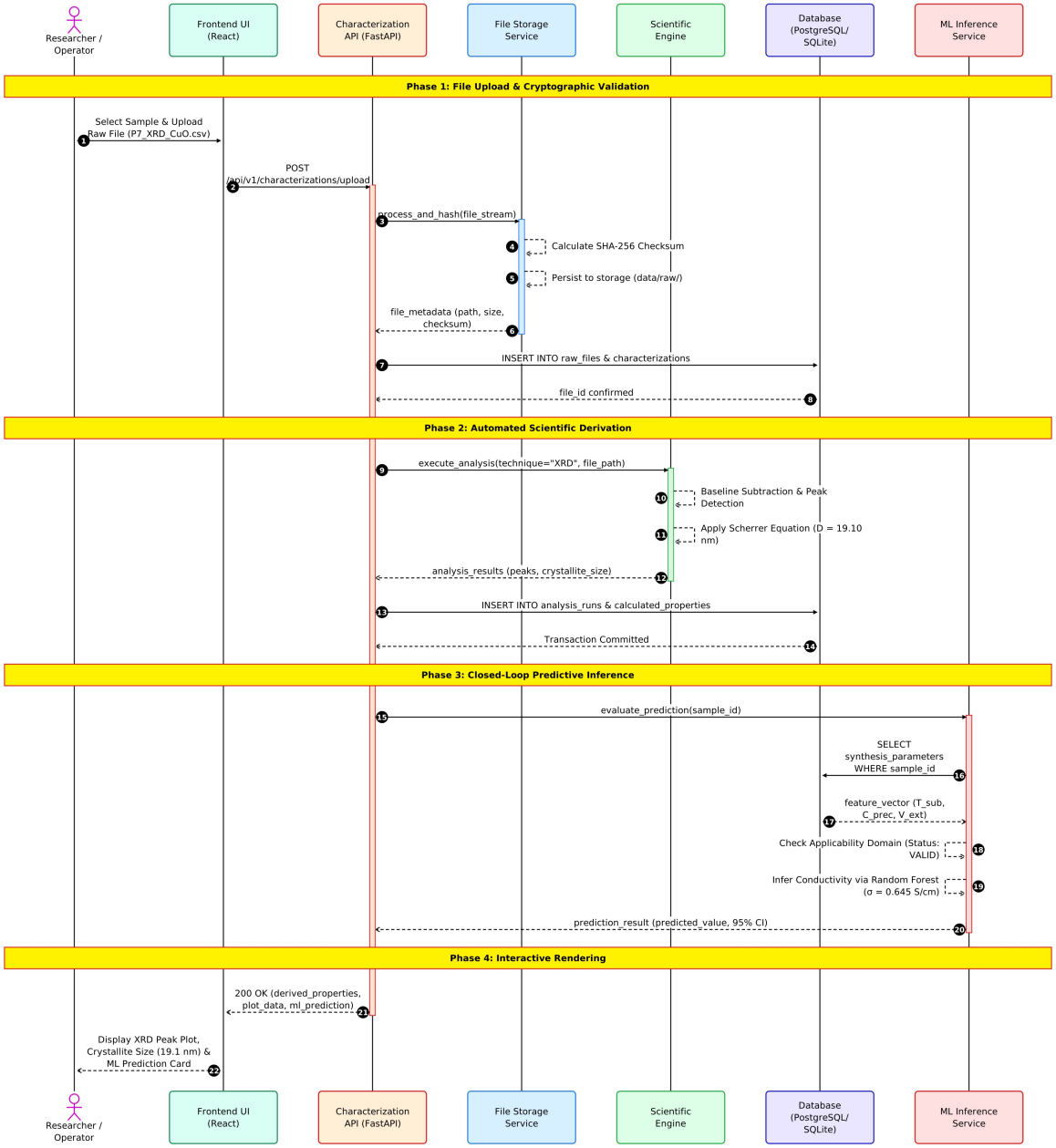


Figure 4.9: UML Sequence Diagram Illustrating End-to-End Analysis and Inference Execution

3. **Phase 3: Closed-Loop Predictive Inference (Steps 15–20):**
    - The API calls `evaluate_prediction(sample_id)` on the ML Inference Service.
    - The service fetches synthesis variables ($T_{\text{sub}} = 350°\text{C}$, $C_{\text{prec}} = 0.05$ M, $V_{\text{ext}} = 10$ mL) from the database.
    - The Applicability Domain Guard verifies that the query point lies within the training bounding box (`Status: VALID`).
    - The Champion Random Forest Regressor predicts electrical conductivity ($\sigma = 0.645$ S/cm) with its 95% confidence interval.
4. **Phase 4: Interactive Rendering (Steps 21–22):**
    - The API responds with an HTTP 200 OK containing parsed spectral plots, calculated properties, and the ML prediction card.
    - The React UI renders the interactive XRD diffractogram, displays the crystallite size banner, and presents the conductivity prediction card.

4.4 Data Dictionary

The Data Dictionary provides complete schema specifications for the GreenSynth Analytics relational database, detailing table attributes, data types, nullability, key constraints, and descriptions.

Table 4.2: Data Dictionary: Research Projects and Experimental Setup Tables

Attribute Name	Data Type	Constraint	Description
Table: `projects`			
`id`	UUID	PK, Not Null	Unique project identifier
`project_code`	VARCHAR(50)	Unique, Not Null	Alphanumeric identifier (e.g., P7-CUO-2026)
`material`	VARCHAR(100)	Not Null	Synthesized material (e.g., Copper Oxide CuO)
`extract`	VARCHAR(100)	Not Null	Phytochemical source (e.g., Morus alba)
`solvent`	VARCHAR(50)	Not Null	Extraction solvent (e.g., Ethanol)
`synthesis_method`	VARCHAR(100)	Not Null	Deposition method (Spray Pyrolysis)
Table: `parameter_definitions`			
`id`	UUID	PK, Not Null	Unique parameter definition identifier
`project_id`	UUID	FK, Not Null	Reference to parent `projects.id`
`parameter_code`	VARCHAR(50)	Not Null	Canonical parameter token (e.g., SUB_TEMP)
`unit`	VARCHAR(20)	Not Null	Measurement unit (e.g., deg C, mL/min, M)
`min_val`	FLOAT	Nullable	Minimum admissible operating bound
`max_val`	FLOAT	Nullable	Maximum admissible operating bound
Table: `experiments`			
`id`	UUID	PK, Not Null	Unique experiment run identifier
`project_id`	UUID	FK, Not Null	Reference to parent `projects.id`
`experiment_code`	VARCHAR(50)	Unique, Not Null	Run code (e.g., EXP-P7-001)
`status`	VARCHAR(30)	Not Null	Run lifecycle state (PLANNED, COMPLETED)
`experiment_date`	DATE	Not Null	Execution calendar date

Table 4.3: Data Dictionary: Experimental Parameters, Samples, and Raw Instrument Files

Attribute Name	Data Type	Constraint	Description
Table: `experiment_parameters`			
`id`	UUID	PK, Not Null	Unique experiment parameter identifier
`experiment_id`	UUID	FK, Not Null	Reference to `experiments.id`
`definition_id`	UUID	FK, Not Null	Reference to `parameter_definitions.id`
`value_numeric`	FLOAT	Nullable	Numerical recorded value (e.g., 350.0)
`value_text`	VARCHAR(255)	Nullable	Qualitative descriptor if non-numeric
Table: `samples`			
`id`	UUID	PK, Not Null	Unique physical sample identifier
`experiment_id`	UUID	FK, Not Null	Reference to parent `experiments.id`
`sample_code`	VARCHAR(50)	Unique, Not Null	Physical specimen code (e.g., P7-CUO-S1)
`material`	VARCHAR(100)	Not Null	Target thin-film material designation
Table: `characterizations`			
`id`	UUID	PK, Not Null	Unique characterization run identifier
`sample_id`	UUID	FK, Not Null	Reference to `samples.id`
`technique`	VARCHAR(50)	Not Null	Technique (XRD, UV-VIS, FTIR, ELEC_IV)
`status`	VARCHAR(30)	Not Null	Processing status (UPLOADED, ANALYZED)
Table: `raw_files`			
`id`	UUID	PK, Not Null	Unique raw file record identifier
`characterization_id`	UUID	FK, Not Null	Reference to parent `characterizations.id`
`checksum_sha256`	VARCHAR(64)	Not Null	Cryptographic SHA-256 file checksum
`storage_path`	VARCHAR(500)	Not Null	File vault storage path on server disk

Table 4.4: Data Dictionary: Scientific Derivations and Machine Learning Model Registry

Attribute Name	**Data Type**	**Constraint**	**Description**
Table: `analysis_runs`			
`id`	UUID	PK, Not Null	Unique algorithmic run identifier
`characterization_id`	UUID	FK, Not Null	Reference to `characterizations.id`
`algorithm_name`	VARCHAR(100)	Not Null	Executed routine (e.g., Scherrer_v1.2)
`status`	VARCHAR(30)	Not Null	Computation status (SUCCESS, FAILED)
Table: `calculated_properties`			
`id`	UUID	PK, Not Null	Unique property identifier
`sample_id`	UUID	FK, Not Null	Reference to `samples.id`
`analysis_run_id`	UUID	FK, Not Null	Reference to generating `analysis_runs.id`
`property_name`	VARCHAR(100)	Not Null	Metric name (e.g., Crystallite Size, Eg, Conductivity)
`value`	FLOAT	Not Null	Derived scalar value (e.g., 19.10, 1.74, 0.645)
`unit`	VARCHAR(30)	Not Null	Physical unit (e.g., nm, eV, S/cm)
Table: `ml_models`			
`id`	UUID	PK, Not Null	Unique ML model identifier
`model_type`	VARCHAR(50)	Not Null	Regressor architecture (RandomForest, XGBoost)
`status`	VARCHAR(30)	Not Null	Deployment status (CANDIDATE, CHAMPION)
`metrics_json`	JSONB / TEXT	Not Null	Serialized performance metrics (R^2, RMSE, MAE)
`artifact_path`	VARCHAR(500)	Not Null	Serialized model weights path (.joblib)

Chapter 5

IMPLEMENTATION

5.1 Technologies Used

GreenSynth Analytics is engineered as a modern, full-stack, modular materials informatics platform designed for high data integrity, automated physical property derivations, and machine-learning benchmarking. The system is partitioned into a decoupled architecture comprising an interactive client-side Single-Page Application (SPA), a high-throughput asynchronous backend server, an enterprise relational database management system, and an integrated scientific computation and machine-learning engine.

5.1.1 Frontend Architecture Stack

The client-side user interface is engineered to provide an intuitive, responsive, and low-latency environment for laboratory researchers, principal investigators, and students.

- **React (v18.2.0):** Serves as the core declarative user interface library. React's component-driven architecture facilitates the isolation of complex laboratory interfaces—such as interactive diffractogram canvases, experiment parameter tables, and tabular upload consoles—into modular, reusable UI components. React 18's concurrent rendering engine guarantees fluid interface updates even during intensive live rendering of multi-thousand-point spectral datasets.

- **TypeScript (v5.3.3):** Enforces strict static typing across the entire frontend application. Custom TypeScript interfaces mirror backend Pydantic models (e.g., parameter limits, characterization metadata, analysis run outputs), eliminating runtime null-pointer exceptions and ensuring compile-time contract safety between client and server.

- **Vite (v5.0.12):** Functions as the frontend build tool and local development environment. Leveraging native ES modules in the browser, Vite delivers sub-second Hot Module Replacement (HMR) and performs Rollup-based production bundling with automated code-splitting and tree-shaking for minimal initial asset transfer.

- **React Router (v6.22.0):** Manages client-side declarative routing and protected navigation hierarchies. It provides route-level authentication guards (such as `AdminRoute` and `ProtectedRoute`) that prevent unauthorized access to sensitive administrative, project-configuration, and model-training views while preserving browser history and bookmarkable states.

- **Axios (v1.6.7):** Acts as the promise-based HTTP client communicating with backend REST endpoints. Axios request interceptors automatically attach the JSON Web Token (JWT) Bearer authorization headers to outgoing requests,

while response interceptors centralize global error handling, authentication expiry redirects, and multipart form-data streaming for multi-megabyte scientific files.

- **Lucide React (v0.323.0) and Recharts (v2.10.4):** Lucide React provides an extensive set of clean, scalable SVG icons representing scientific instruments, file formats, and operational states. Recharts delivers declarative, responsive SVG charting components capable of rendering high-density XRD diffractograms, UV-Vis absorbance spectra, Tauc tangent extrapolations, and Keithley I–V curves with synchronized tooltips, crosshairs, and dynamic zoom windows.
- **Tailwind CSS (v3.4.1):** Provides utility-first responsive styling utilizing a dark-mode scientific aesthetic. A curated palette featuring slate neutrals, emerald accents, and indigo interactive highlights minimizes optical fatigue during prolonged laboratory data entry and spectral analysis.

5.1.2 Backend Architecture Stack

The server-side application is implemented in Python 3.11, structured around high-throughput asynchronous execution, strict schema enforcement, and deterministic scientific processing.

- **Python (v3.11):** Selected as the core server language due to its native support for advanced scientific libraries, optimized C-extensions, and enhanced asynchronous execution performance via the updated Python 3.11 bytecode compiler.
- **FastAPI (v0.109.0):** Provides the core asynchronous RESTful application framework. Built on Starlette and AnyIO, FastAPI natively processes concurrent I/O operations (file streaming, multi-query database calls) via Python's `asyncio` event loop. It automatically generates interactive OpenAPI 3.0 (Swagger) and ReDoc documentation directly from source code annotations.
- **SQLAlchemy (v2.0.25):** Serves as the Object-Relational Mapping (ORM) and SQL expression engine. Configured with modern 2.0-style asynchronous session scopes (`AsyncSession`), it guarantees strict ACID compliance, managed transactional boundaries, typed relational queries, and automated foreign-key cascade enforcement.
- **Pydantic (v2.6.0):** Powers runtime schema validation, data serialization, and configuration management. Pydantic's Rust-accelerated core validates incoming HTTP request bodies against physical science boundaries (e.g., verifying that synthesis temperatures, concentrations, and pH values reside within allowable physical limits) before database persistence.
- **Asynchronous Database Connectivity (`asyncpg`):** A high-performance, non-blocking PostgreSQL driver executing native binary protocol communications. Integrated with SQLAlchemy's asynchronous engine, it supports connection pooling and high-concurrency request handling during batch scientific data uploads.

5.1.3 Database Infrastructure

The platform employs a robust relational database layer configured to support both enterprise production deployments and rapid, zero-friction local development:

- **PostgreSQL (v16.1):** Selected as the primary persistent database for production and staging environments. PostgreSQL offers enterprise-grade reliability, foreign-key constraint enforcement, full-text search, and native JSONB data types used for semi-structured instrument configuration headers and dynamic experiment parameters.
- **SQLite (v3.42):** Maintained with full compatibility within SQLAlchemy models to allow lightweight, zero-configuration local development and rapid, isolated unit testing under automated testing suites without external service dependencies.

5.1.4 Scientific Computing, Analytics, and Machine Learning Stack

The analytical engine integrates the standard Python scientific stack, ensuring deterministic mathematical calculations and reproducible regression modeling:

- **NumPy (v1.26.3):** Provides optimized multi-dimensional array structures and low-level vectorized mathematical routines for rapid processing of dense coordinate arrays (2θ–I, λ–A, V–I).
- **Pandas (v2.2.0):** Powers tabular data manipulation, multi-column ASCII instrument file parsing, delimiter inference (comma, tab, space), missing-value sanitization, and dataset feature extraction.
- **SciPy (v1.12.0):** Provides digital signal processing and numerical optimization routines. Specifically, the `scipy.signal` module executes Savitzky-Golay polynomial smoothing and baseline estimation, while `scipy.optimize` executes non-linear least-squares curve fitting (Levenberg-Marquardt algorithm) for XRD peak profile modeling.
- **Scikit-learn (v1.4.0):** Powers the machine-learning pipeline, providing data preprocessing (`StandardScaler`), cross-validated regression models (Random Forest, Gradient Boosting, Ridge Regression), and comprehensive evaluation metrics (R^2, RMSE, MAE).
- **Matplotlib (v3.8.2) and Plotly (v5.18.0):** Supply publication-quality static and interactive charting. Matplotlib renders vector graphics incorporated directly into downloadable dossiers, while Plotly provides web-ready interactive JSON serialization for dynamic client inspection.
- **ReportLab (v4.0.9):** Facilitates programmatic PDF generation, compiling experimental metadata, characterization diffractograms, optical plots, and derived material properties into standardized, publication-grade research dossiers.

5.1.5 Database Migration Framework

- **Alembic (v1.13.1):** Acts as the database schema migration environment. Alembic tracks schema changes under Git version control, automatically generating migration scripts from SQLAlchemy model updates, verifying database migration integrity, and supporting reproducible forward and backward rollbacks across production and development databases.

5.1.6 Deployment, Hosting, and Supporting Infrastructure

- **Containerization and Deployment Configuration:** The application is containerized using Docker and orchestrated via Docker Compose, defining separate containers for the FastAPI application server, PostgreSQL database, and frontend build proxy.
- **ASGI Server:** Production serving is powered by Uvicorn running behind an Nginx reverse proxy configured for SSL/TLS termination, HTTP/2 multiplexing, static asset caching, and request rate limiting.
- **File Storage Infrastructure:** Uploaded raw characterization files are managed via a dedicated, secure local filesystem storage abstraction. Files are organized by their SHA-256 cryptographic digest under partitioned directory structures (e.g., `storage/raw/prefix/checksum`), mitigating filesystem directory saturation and preventing path-traversal vulnerabilities.
- **Security and Cryptographic Services:** User passwords are encrypted using Argon2/bcrypt adaptive hashing. API interactions are secured via OAuth2 Password Bearer authentication emitting signed HMAC-SHA256 (HS256) JSON Web Tokens with configurable validity windows.

5.2 Algorithm

5.2.1 Data Management, Validation, and Transformation Procedures

GreenSynth Analytics enforces a deterministic data processing pipeline that ensures data integrity from raw laboratory instrument ingestion through to finalized physical property derivation and predictive modeling.

1. **Input Validation and Boundary Enforcement:** When a researcher logs synthesis parameters (e.g., precursor molarity, solvent ratio, spray pyrolysis substrate temperature), the data payload is validated against strict Pydantic schemas. If a parameter exceeds physical boundaries (e.g., substrate temperature < 100°C or > 600°C), the transaction is rejected with descriptive error payloads.

2. **Cryptographic Hashing and Deduplication:** Uploaded characterization files are processed via a streaming SHA-256 hashing algorithm. If a file with an identical hash already exists within the target project scope, the system flags the upload to avoid redundant storage and prevent dataset duplication.

3. **Instrument Header Sniffing and Sanitization:** Raw instrument ASCII data are inspected line-by-line. The parser identifies variable-length metadata header lines (instrument model, scan rate, voltage, tube type), determines delimiters (comma, tab, whitespace), scrubs non-numeric tokens, and parses numeric arrays into normalized coordinate vectors.

4. **Signal Conditioning and Background Subtraction:** Raw spectral data undergo digital signal conditioning. A Savitzky-Golay filter removes high-frequency detector scintillation without compromising legitimate peak morphology. Baseline drift caused by substrate scattering is modeled and subtracted using asymmetric least squares.

5. **Physical Property Extraction:** Domain-specific algorithms extract material properties (crystallite size, optical bandgap, electrical conductivity), which are validated against expected semiconductor bounds and persisted to the relational database.

5.2.2 Core Scientific Derivation Algorithms

The platform implements five primary computational algorithms:

Algorithm 1: Cryptographic Ingestion and Tabular Stream Parsing

1. **Stream Hashing:** As bytes are streamed from the client, a running SHA-256 digest is accumulated:

$$\text{digest} = \mathcal{H}_{\text{SHA-256}}(\text{raw_byte_stream}) \tag{5.1}$$

2. **Deduplication Check:** Query database index on `raw_files.checksum_sha256`. If matching record exists, attach reference or raise collision warning.

3. **Delimiter and Header Detection:** Read leading lines to detect instrument metadata tokens. Extract numeric columns into coordinate array $\mathbf{D} = \{(x_i, y_i)\}_{i=1}^{N}$.
4. **Sanitization:** Filter out non-numeric values, NaN tokens, and sensor over-range artifacts.

Algorithm 2: Automated XRD Scherrer Crystallite Size Derivation

The routine extracts average crystallite size (D) along dominant diffraction planes using the Scherrer formula:

$$D = \frac{K\lambda}{\beta \cos\theta} \tag{5.2}$$

where $K = 0.94$ is the crystallite shape factor, $\lambda = 0.15406$ nm (Cu-$K\alpha$ radiation), θ is the Bragg diffraction angle in radians, and β is the Full Width at Half Maximum (FWHM) in radians, corrected for instrumental broadening:

$$\beta = \sqrt{\beta_{\text{observed}}^2 - \beta_{\text{instrumental}}^2} \tag{5.3}$$

Execution Workflow:

1. Apply Savitzky-Golay digital filter (window length = 7, polynomial order = 2) to smooth raw counts without peak distortion.
2. Identify peak maxima where $\frac{dI}{d(2\theta)} = 0$ and $\frac{d^2I}{d(2\theta)^2} < 0$, filtering candidates by prominence threshold ($P \geq 0.15 \times I_{\max}$).
3. Fit peak profiles—specifically the monoclinic CuO ($\bar{1}11$) and (111) reflections at $2\theta \approx 35.5^\circ$ and 38.7°—with a Pseudo-Voigt distribution to extract centroid $2\theta_0$ and β.
4. Evaluate Equation (5.2) and store derived crystallite size in `calculated_properties`.

Algorithm 3: Automated UV-Vis Tauc Optical Bandgap Extraction

For direct allowed interband transitions in semiconducting CuO, the optical absorption coefficient α near the absorption edge obeys Tauc's relation:

$$(\alpha h\nu)^2 = A(h\nu - E_g) \tag{5.4}$$

where $h\nu = \frac{1239.8}{\lambda\ (\text{nm})}$ represents photon energy in eV, A is an energy-independent constant, and E_g is the optical bandgap. **Execution Workflow:**

1. Convert experimental wavelength λ and absorbance A_{abs} into photon energy $h\nu$ and $(\alpha h\nu)^2$.
2. Compute numerical first derivative $\frac{d}{d(h\nu)}[(\alpha h\nu)^2]$ to identify the maximum slope coordinate (inflection point $h\nu_{\text{infl}}$).
3. Fit a linear regression line $y = m(h\nu) + c$ over the linear absorption window surrounding $h\nu_{\text{infl}}$.
4. Determine the x-intercept at $(\alpha h\nu)^2 = 0$, yielding the direct bandgap:

$$E_g = -\frac{c}{m} \quad \text{(eV)} \tag{5.5}$$

Algorithm 4: Steady-State Electrical I–V Ohmic Regression and Conductivity Derivation

1. Ingest voltage-current coordinate vectors (V_i, I_i) from Keithley 2450 SourceMeter scans across the linear sweep window $[-10\text{ V}, +10\text{ V}]$.

2. Fit linear regression:
$$I = \left(\frac{1}{R}\right) V + I_{\text{offset}} \tag{5.6}$$
Evaluate $R^2 \geq 0.998$ to confirm Ohmic behavior.

3. Calculate film electrical conductivity σ using electrode length L, gap W, and thickness $t \approx 400$ nm:
$$\sigma = \frac{L}{R \cdot W \cdot t} \quad (\text{S/cm}) \tag{5.7}$$

Algorithm 5: Machine Learning Predictive Modeling & Cross-Validation

1. Assemble feature vectors $\mathbf{x} = [T_{\text{sub}}, M, C_{\text{ext}}, Q, \text{pH}]$ from `experiment_parameters`.

2. Target vectors $\mathbf{y}$ include bandgap (E_g), crystallite size (D), and conductivity (σ).

3. Apply 5-fold cross-validation across Random Forest and XGBoost regressors, tracking R^2, RMSE, and MAE across training and validation folds.

4. Store model weights and validation metrics in `ml_models` and `ml_training_runs`.

5.2.3 Physical Materials Synthesis Implementation

To validate the digital informatics pipeline with real laboratory data, experimental synthesis was conducted investigating phytochemical-assisted CuO thin films deposited via chemical spray pyrolysis.

Precursor Chemistry and Stoichiometric Calculations

Analytical reagent grade copper sulfate pentahydrate ($CuSO_4{\cdot}5H_2O$, $M_r = 249.68$ g/mol) was reacted with sodium hydroxide (NaOH, $M_r = 40.00$ g/mol) in the presence of filtered *Morus alba* (mulberry) leaf extract in ethanol:

$$CuSO_4(aq) + 2NaOH(aq) \xrightarrow{\text{phytochemicals}} Cu(OH)_2(s) \downarrow + Na_2SO_4(aq) \tag{5.8}$$

$$Cu(OH)_2(s) \xrightarrow{\Delta} CuO(s) + H_2O(g) \uparrow \tag{5.9}$$

In a representative laboratory trial, 7.49 g of $CuSO_4 \cdot 5H_2O$ was dissolved in 75 mL deionized water (0.40 M), combined with 75 mL of mulberry leaf extract, and titrated with 2.40 g NaOH in 60 mL water (1.00 M) at 70°C for 30 min. The monitored pH dynamic trajectory evolved from 2.45 (pure $CuSO_4$), to 2.60 upon extract addition, reaching 12.36 upon complete titration.

Chemical Spray Pyrolysis Deposition Envelope

The stabilized precursor solution was atomized onto pre-cleaned borosilicate glass substrates held at 350°C–400°C using clean compressed air carrier gas (1.8 bar) at a liquid flow rate of 3.5 mL/min over a nozzle-substrate distance of 28.0 cm. Deposition proceeded for 25 min with a 10 s ON/15 s OFF duty cycle, preventing thermal quenching and producing adherent, nanocrystalline CuO thin films.

5.2.4 Workflow

The platform realizes a unified, 12-stage end-to-end research workflow, depicted structurally below:

1. **Stage 1: User Authentication and Access Authorization:** Researchers authenticate via JWT-secured credentials. The system establishes user identity, assigned research groups, and role permissions.

2. **Stage 2: Research Project Selection:** The authenticated user selects an active research project (e.g., `P7-CUO-2026`). Project configuration catalogs (allowable precursors, biomass, deposition methods) are loaded into context.

3. **Stage 3: Definition and Recording of Synthesis Parameters:** Researchers configure experimental run parameters (temperatures, concentrations, flow rates, stoichiometric ratios) validated against admissible physical bounds.

4. **Stage 4: Creation of Experiments and Associated Samples:** The platform persists the experiment record and generates unique sample identifiers (e.g., `P7-CUO-S01`) associated with specific substrates and deposition runs.

5. **Stage 5: Upload and Storage of Characterization Files:** Raw characterization files (XRD, UV-Vis, FTIR, SEM, I–V) are uploaded. The system calculates running SHA-256 hashes, detects duplicates, and stores raw streams in partitioned storage.

6. **Stage 6: Processing of Experimental Data:** The tabular ingestion engine sniffs file headers, removes non-numeric tokens, isolates coordinate data, and applies Savitzky-Golay signal smoothing.

7. **Stage 7: Extraction of Relevant Scientific Features:** Peak locations, FWHM values, absorption edges, and current-voltage slopes are extracted from conditioned datasets.

8. **Stage 8: Calculation and Storage of Material Properties:** Physical property engines evaluate Scherrer crystallite sizes, Tauc optical bandgaps, and Ohmic electrical conductivities, persisting results to `calculated_properties`.

9. **Stage 9: Preparation of Datasets for Statistical Analysis and Machine Learning:** Validated experimental features and calculated properties are compiled into structured tabular ML datasets, with explicit partitioning between training and validation sets.

10. **Stage 10: Model Training, Evaluation, and Prediction:** Machine learning algorithms (Random Forest, XGBoost) are trained and cross-validated. Performance metrics (R^2, RMSE, MAE) are recorded, and trained models are deployed for property prediction.

11. **Stage 11: Design of Experiments (DOE) and Candidate Generation (Optional Stage):** *Optional/Exploratory Stage:* The platform employs factorial or Latin Hypercube sampling to recommend candidate experimental parameters aimed at target property optimization.

12. **Stage 12: Review of Results and Laboratory Validation:** Researchers inspect dashboards, evaluate ML predictions against physical bench results, and export comprehensive PDF dossiers to close the experimental loop.

5.3 Database Design

5.3.1 Database Architecture and Entity-Relationship Model

The GreenSynth Analytics database is architected in Third Normal Form (3NF) to guarantee referential integrity, eliminate data redundancy, and maintain strict traceability from raw instrument files back to experimental synthesis runs.

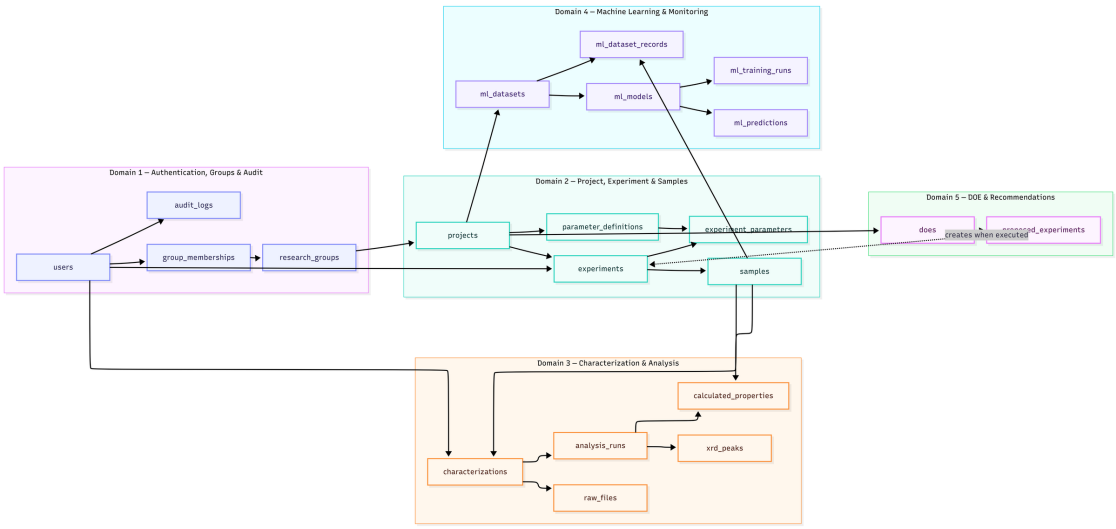


Figure 5.1: Entity-Relationship (ER) Architecture of GreenSynth Analytics: Detailed relational model illustrating core tables, primary/foreign key mappings, and cascading integrity constraints.

5.3.2 Entities, Schemas, and Relational Structure

The database architecture is structured across eight functional relational domains:

1. User Accounts, Roles, and Research Group Membership

- `users`: Stores researcher credentials, full name, hashed passwords (Argon2/bcrypt), email address, system role (`admin`, `researcher`, `student`, `viewer`), active status,

and audit timestamps.

- `research_groups`: Defines laboratory research teams, principal investigator ownership, and institutional affiliations.
- `group_memberships`: Associative table mapping users to research groups with group-specific roles and invitation tracking.
- `invitations`: Manages pending, accepted, or expired research group invitation tokens.

2. Research Projects and Configuration Catalogs

- `projects`: Top-level organizational entity storing project code (e.g., `P7-CUO-2026`), title, description, research group foreign key, and lifecycle state (`active`, `archived`).
- `project_definitions`: Defines project-specific parameter bounds, target material families, and operational constraints.
- `materials_catalog`, `biomass_catalog`, `solvents_catalog`, `synthesis_methods_catalog`: Standardized reference catalogs defining acceptable botanical extracts, precursor salts, solvents, and deposition techniques.

3. Experiments, Parameters, and Samples

- `experiments`: Records individual synthesis runs, linking to parent project, creator ID, run status (`PLANNED`, `COMPLETED`, `FAILED`), and run timestamps.
- `parameter_definitions`: Defines expected experimental variables (e.g., `substrate_temperat precursor_molarity`), measurement units, and admissible minimum/maximum bounds.
- `experiment_parameters`: Key-value relational entity storing actual recorded synthesis values for an experiment, enforcing composite uniqueness on `(experiment_id, parameter_definition_id)`.
- `samples`: Represents physical fabricated specimens resulting from an experiment, storing unique sample codes (e.g., `P7-CUO-S01`), storage locations, substrate type, and creation date.

4. Characterization Records and Raw Files

- `characterizations`: Associates a physical sample with an analytical measurement event, recording instrument type (`XRD`, `UV_VIS`, `FTIR`, `SEM`, `ELECTRICAL`), operator, and timestamp.
- `raw_files`: Stores metadata for ingested instrument files, including original filename, MIME type, file size, SHA-256 cryptographic checksum, and storage file path. Enforces unique constraint on `checksum_sha256`.

5. Analysis Runs, Processed Files, and Extracted Features

- `analysis_runs`: Tracks execution of scientific derivation algorithms, storing routine name (e.g., `scherrer_crystallite_size`), execution status, input parameters, and runtime duration.
- `processed_files`: References conditioned, filtered coordinate datasets generated from raw files.
- `xrd_peaks`: Stores individual fitted diffraction peaks ($2\theta_0$, FWHM β, intensity I, Miller indices (hkl)).
- `ftir_annotations`, `sem_metadata`, `sem_measurements`: Stores domain-specific annotations including infrared vibrational band assignments and SEM grain boundary diameter measurements.

6. Calculated Material Properties

- `calculated_properties`: Centralized repository storing standardized physical metrics derived from analysis runs. Records property name (e.g., `optical_bandgap`, `crystallite_size`, `electrical_conductivity`), numeric value, unit of measure, measurement uncertainty, and algorithm reference.

7. Machine Learning Datasets, Models, and Predictions

- `ml_datasets`: Defines curated feature-target matrices assembled for machine learning.
- `ml_dataset_records`: Associative table linking specific samples and calculated properties to an ML dataset.
- `ml_training_runs`: Tracks training job parameters, train/test split ratios, and cross-validation configurations.
- `ml_models`: Stores serialized model artifacts (weights, pipelines), algorithm type (Random Forest, XGBoost), and validation scores (R^2, RMSE, MAE).
- `ml_predictions`: Records model inference outputs for hypothetical synthesis parameter configurations.
- `prediction_validations`: Maps predicted material properties against subsequent physical laboratory measurements, tracking prediction error and model health.

8. Experimental Design (DOE) and Recommendations

- `does`, `proposed_experiments`, `doe_analyses`: Manages Design of Experiments exploration matrices and candidate synthesis conditions.
- `recommendations`, `optimization_runs`: Stores algorithmic candidate parameter suggestions generated via multi-objective Bayesian or factorial optimization.

5.3.3 Constraints, Data Integrity, and Access Security

- **Referential Integrity:** Foreign keys employ explicit cascading actions (`ON DELETE CASCADE` for dependent records such as parameters and analysis runs; `ON DELETE RESTRICT` for core user and project definitions).
- **Cryptographic Deduplication:** The unique constraint on `raw_files.checksum_sha256` prevents identical characterization files from being stored multiple times.
- **Parameter Bounds Constraints:** Check constraints validate numeric ranges (e.g., substrate temperature $T_{\text{sub}} > 0$, solution pH $\in [0, 14]$).
- **Synthetic vs. Experimental Data Segregation:** The schema explicitly incorporates an `is_synthetic` boolean flag across experimental and dataset records. This guarantees that synthetic benchmark suites (used for stress-testing ML pipelines) are never conflated with physical laboratory measurements.
- **Audit Logging:** The `audit_logs` table captures an immutable record of all data modifications, logging user ID, timestamp, IP address, target table, and JSON diffs of modified fields.

5.4 Modules Description

5.4.1 User Authentication and Research Access Module

- **Purpose:** Governs secure platform entry, identity verification, role-based authorization, and project-level data segregation.
- **Inputs:** User credentials (email, password), invitation tokens, role assignment requests.
- **Processing Logic:** Passwords are verified against stored hashes using Argon2/bcrypt. Upon successful authentication, the module issues signed HMAC-SHA256 JWT access tokens containing user claims, role privileges, and expiration times. Endpoint dependency guards inspect tokens on every incoming request, verifying group membership and project ownership before granting read/write permissions.
- **Outputs:** Signed JWT bearer tokens, user profile contexts, authorization rejection exceptions (HTTP 401/403).
- **Interactions:** Interacts directly with all downstream modules by injecting verified user and project context into every API request.

5.4.2 Experimental Data and Characterization Management Module

- **Purpose:** Provides comprehensive lifecycle management for synthesis experiments, physical samples, and raw characterization files.
- **Inputs:** Synthesis condition parameters, sample metadata, raw ASCII/binary characterization files (XRD, UV-Vis, FTIR, SEM, I–V).

- **Processing Logic:** Validates input parameters against catalog bounds; generates unique sample identifiers; streams uploaded files through a SHA-256 cryptographic hashing pipeline; rejects duplicate files; sniffs column delimiters; cleans numeric coordinates; and writes files to partitioned storage.
- **Outputs:** Persisted experiment records, linked sample entities, sanitized coordinate datasets, raw file audit metadata.
- **Interactions:** Feeds conditioned coordinate arrays into the Scientific Analytics and Machine-Learning Module.

5.4.3 Scientific Analytics and Machine-Learning Module

- **Purpose:** Automates mathematical property derivations, executes cross-validated regression modeling, and generates candidate experimental conditions.
- **Inputs:** Conditioned spectral/diffraction datasets, sample parameter vectors, ML model hyperparameter configurations.
- **Processing Logic:** Executes digital signal processing (Savitzky-Golay smoothing, baseline estimation); performs non-linear curve fitting for Scherrer crystallite sizing; calculates Tauc optical bandgaps via inflection point analysis; derives electrical conductivities from I–V slopes; curates ML training matrices; trains Random Forest and XGBoost regressors under 5-fold cross-validation; and evaluates regression metrics (R^2, RMSE, MAE).
- **Outputs:** Standardized calculated properties, regression model weights, cross-validation metrics, interactive Recharts visualizations, and downloadable PDF dossiers.
- **Interactions:** Consumes data from the Characterization Management Module and writes derived properties back to the central database.

5.5 Screenshots

This section presents authentic graphical interfaces of the implemented GreenSynth Analytics platform across its laboratory management and automated characterization analysis modules, alongside physical laboratory apparatus utilized for experimental synthesis.



Figure 5.2: Executive Dashboard: Overview of active research initiatives, synthesis experiment counts, sample inventories, and recent characterization events.



Figure 5.3: Experiment and Sample Management Interface: Logging synthesis run parameters, solution concentrations, substrate temperatures, and tracking sample life-cycles.

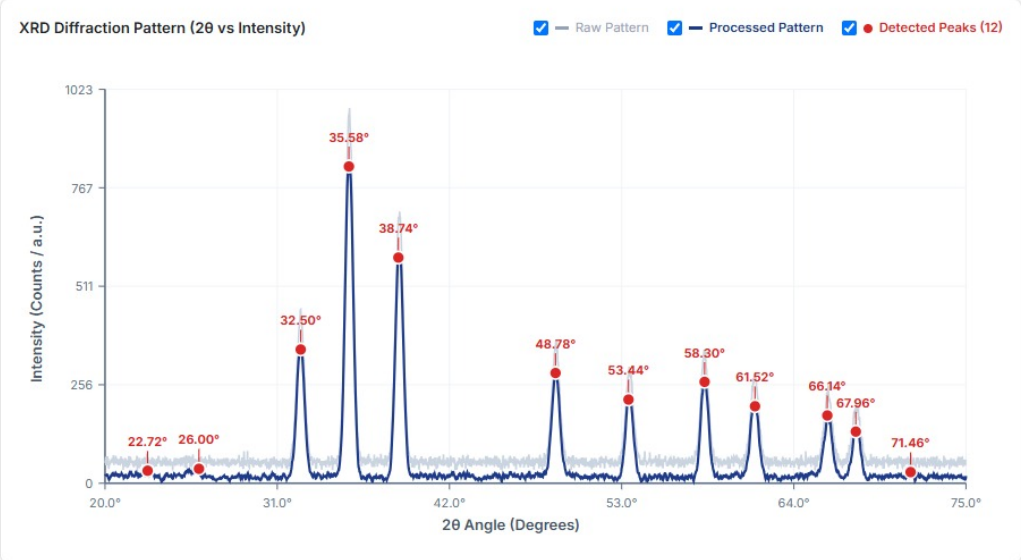


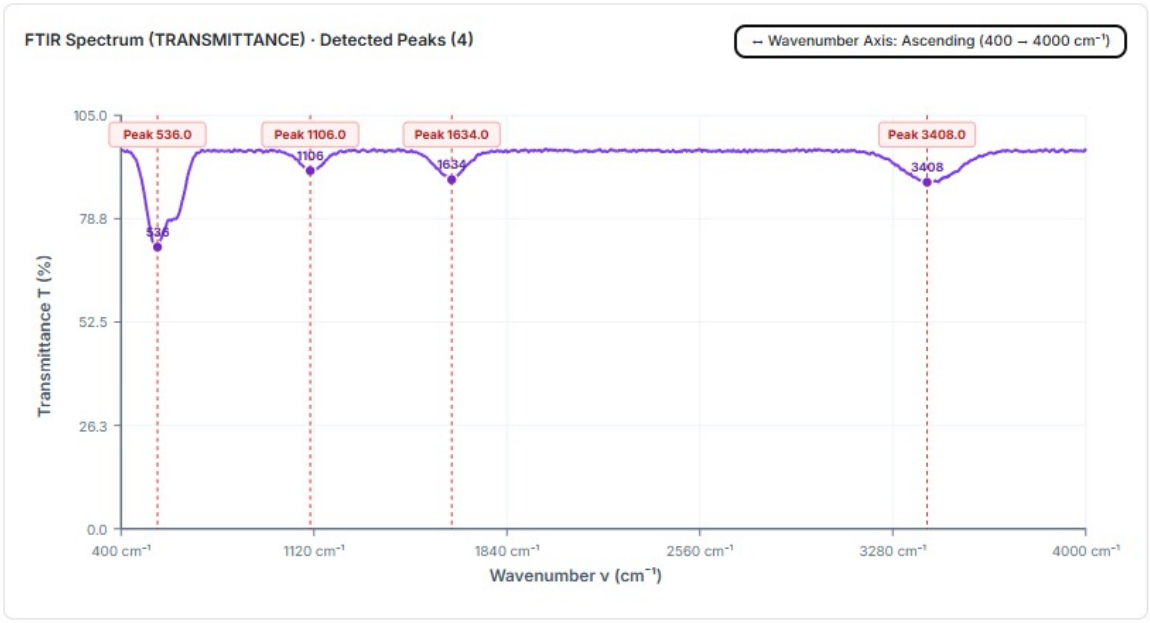
Figure 5.4: X-Ray Diffraction (XRD) Diffractogram Analysis Interface: Interactive 2θ vs. Intensity pattern visualization showing raw background trace, Savitzky-Golay smoothed curve, and automated detection of 12 Bragg reflection peaks (32.50°, 35.58°, 38.74°, 48.78°, 53.44°, 58.30°, 61.52°, 66.14°, 67.96°, 71.46°).



Figure 5.5: UV-Visible Optical Absorbance Spectrum Interface: Experimental absorbance curve (A vs. λ) across 300 nm–800 nm revealing the fundamental interband electronic absorption edge onset of synthesized nanostructured copper oxide.



Figure 5.6: Automated Tauc Plot Optical Bandgap Extrapolation Interface: Direct allowed transition ($n = 2$) $(A \cdot h\nu)^2$ vs. photon energy $h\nu$ curve with linear regression tangent extrapolation yielding an optical bandgap $E_g = 1.359$ eV ($R^2 = 0.9947$).



Detected Peaks & Researcher Annotations (4)

WAVENUMBER (CM⁻¹)	SIGNAL HEIGHT	PROMINENCE	WIDTH (CM⁻¹)	RESEARCHER ANNOTATION	CONFIDENCE
536.0	71.53	24.59	141.4	Peak 536.0 (Detected FTIR Peak)	High
3408.0	88.00	8.42	242.8	Peak 3408.0 (Detected FTIR Peak)	High
1634.0	88.65	7.74	128.7	Peak 1634.0 (Detected FTIR Peak)	High
1106.0	90.93	5.34	106.2	Peak 1106.0 (Detected FTIR Peak)	High

Figure 5.7: Fourier-Transform Infrared (FTIR) Spectroscopic Analysis Interface: Transmittance spectrum ($T\%$ vs. wavenumber ν from 400 to 4000 cm^{-1}) with automated detection of 4 characteristic vibrational peaks (536.0 cm^{-1} Cu–O lattice vibration, 1106.0 cm^{-1}, 1634.0 cm^{-1}, and 3408.0 cm^{-1} hydroxyl stretching) alongside the researcher functional group annotation panel.

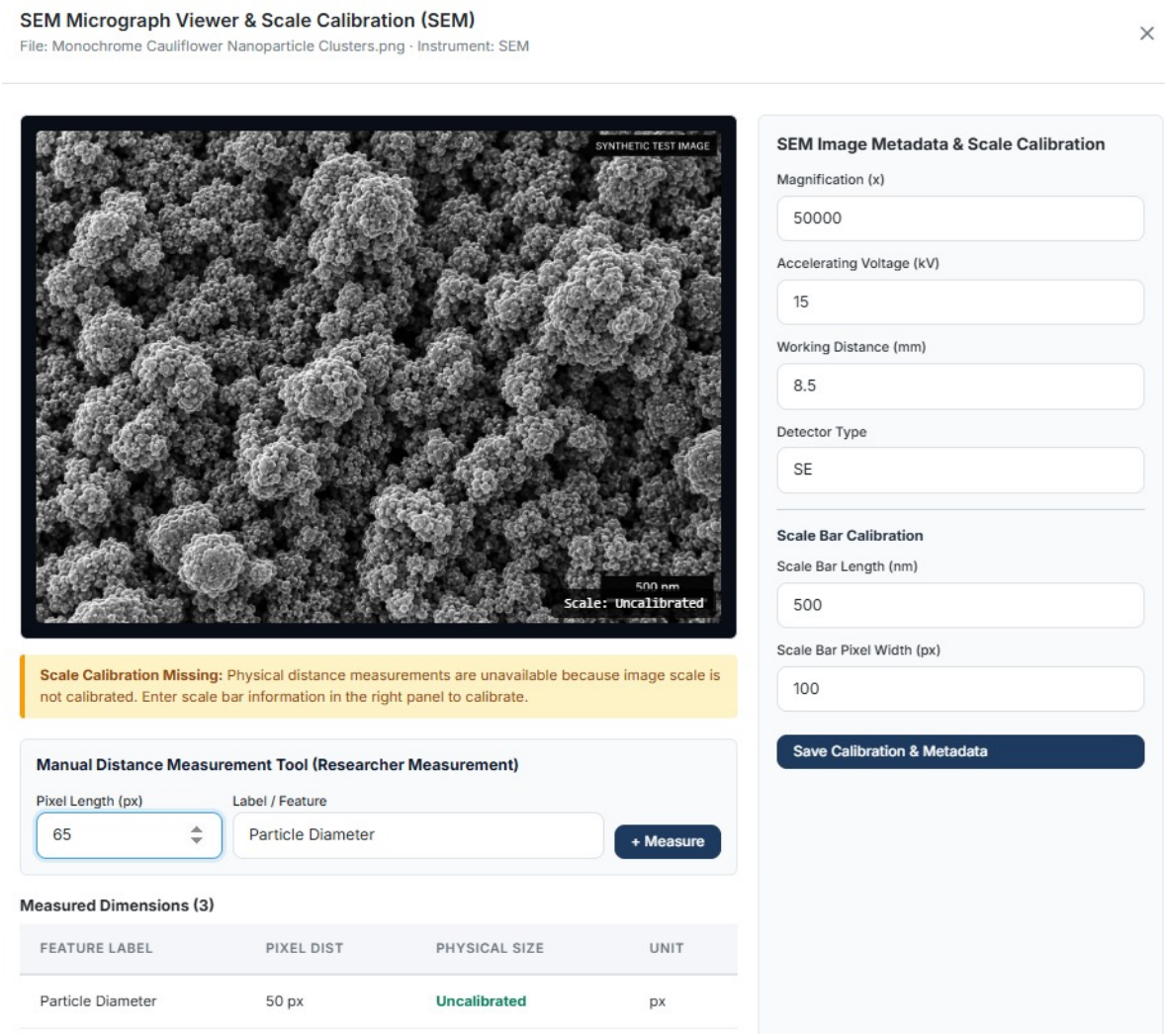


Figure 5.8: Scanning Electron Microscopy (SEM) Micrograph Viewer and Scale Calibration Interface: Micrograph inspection of cauliflower-like nanostructured morphology at $50,000\times$ magnification (15 kV accelerating voltage, 8.5 mm working distance, SE detector) with 500 nm scale bar calibration and interactive digital caliper measurement tool.

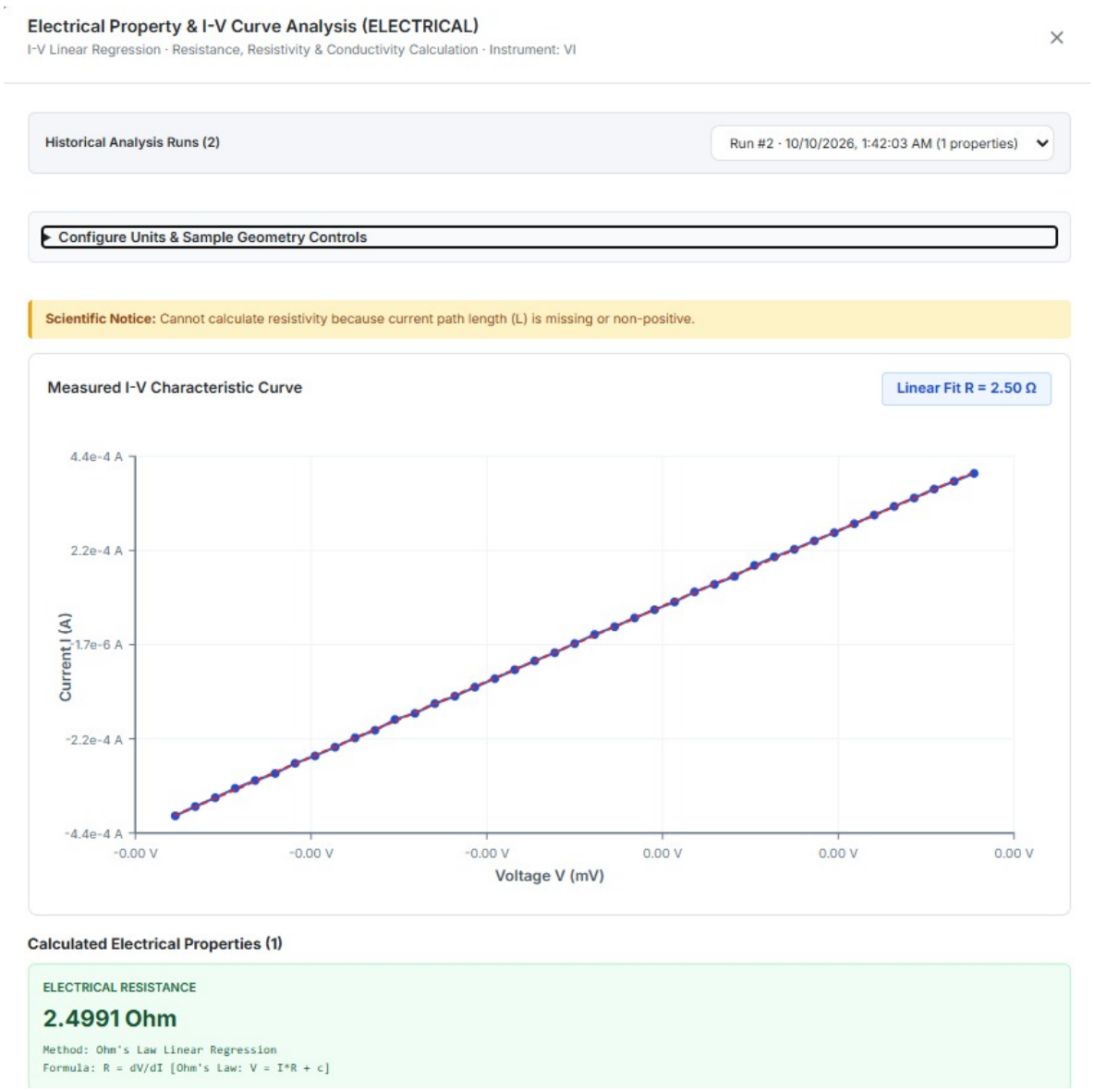


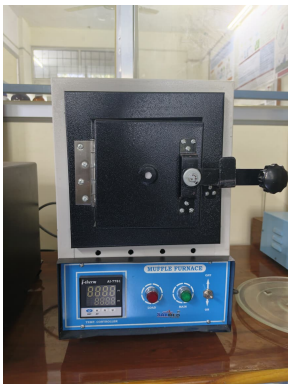
Figure 5.9: Electrical Property and Current-Voltage (I–V) Transport Analysis Interface: Experimental I–V characteristic curve exhibiting Ohmic linearity, linear regression slope ($R = 2.50\,\Omega$), and automated resistance ($2.4991\,\Omega$) and conductivity calculation.



(a) Soxhlet Extraction Setup



(b) Magnetic Stirrer with Hot Plate



(c) Muffle Furnace Chamber

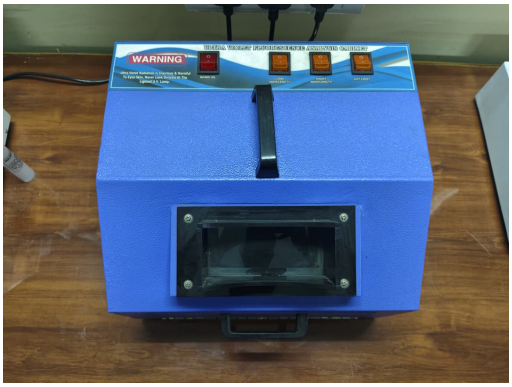


(d) Digital Conductivity Meter

Figure 5.10: Laboratory Precursor Extraction and Synthesis Realization: (a) Solid-liquid Soxhlet reflux apparatus for mulberry leaf extraction, (b) Hot plate magnetic stirrer during precursor compounding, (c) High-temperature muffle furnace for thermal processing, and (d) Digital conductivity meter for electrolyte solution monitoring.



(a) Digital Spectrophotometer



(b) UV Fluorescence Cabinet



(c) Synthesis Glassware and Analytical Balance



(d) Spray Pyrolysis Compressor and Hot Plate

Figure 5.11: Analytical Characterization and Deposition Facilities: (a) Digital spectrophotometer instrumentation, (b) Ultraviolet fluorescence inspection cabinet, (c) Chemical precursor preparation table with precision analytical balance, and (d) Spray pyrolysis pneumatic compressor with substrate hot plate assembly.

Chapter 6

TESTING AND VALIDATION

6.1 Testing Strategy

To ensure high scientific reliability, data integrity, and operational robustness, GreenSynth Analytics was validated using a multi-tiered testing strategy spanning unit, integration, scientific benchmark, and security testing.

- **Unit Testing:** Implemented using `pytest` for backend services and `vitest` for frontend components. Unit tests verify isolated business logic, Pydantic schema boundary constraints, and pure mathematical derivation functions (Savitzky-Golay digital filter, Scherrer crystallite equation, Tauc inflection slope search, and Ohm's law linear regression).
- **Integration Testing:** Validates multi-component workflows, including asynchronous database transactions, multipart characterization file upload pipelines, cryptographic SHA-256 hashing, and automated property derivation triggering upon file ingestion.
- **Scientific Benchmark Validation:** To ensure that automated algorithms yield publishable physical metrics, computed properties were benchmarked against industry-standard analytical software (**OriginLab 2024**) and manual laboratory derivations using authentic P7 experimental datasets (`P7_XRD_CuO.csv`, `P7_UVVIS_CuO.csv`, `P7_ELECTRICAL_CuO.csv`).
- **Security and Access Control Testing:** Verifies stateless JWT token issuance, token expiration handling, Argon2/bcrypt password verification, and role-based access restrictions across `Admin`, `Researcher`, and `Student` accounts.

6.2 Test Cases

This section details the critical test cases executed across the platform. The testing matrix covers functional access control, parameter validation, cryptographic file ingestion, automated scientific calculations, and machine learning model validation.

Table 6.1: Core System, Security, and Data Ingestion Test Cases

Test Case ID	Test Case Objective	Prerequisite	Steps	Input Data	Expected Output	Actual Output	Status
TC-01	Verify user login and JWT token generation	Registered active user in database	1. Enter valid email and password. 2. Submit login form.	Email: `user@lab.in` Pass: `ValidPass123!`	HTTP 200; signed JWT token returned; user routed to dashboard	Token returned; claims verified; redirected	**Pass**
TC-02	Validate synthesis parameter physical boundaries	Active project context (`P7-CUO`)	1. Open parameter form. 2. Enter out-of-bound substrate temp ($T > 600°C$). 3. Submit.	T_{sub} = 750°C (Max allowable: 600°C)	HTTP 422 Unprocessable Entity; descriptive boundary error	Rejection with validation message	**Pass**
TC-03	Prevent duplicate raw file ingestion via SHA-256	Sample record created	1. Upload `P7_XRD.csv`. 2. Attempt uploading identical file to same sample.	Identical byte stream of `P7_XRD.csv`	First upload succeeds; second upload raises duplicate hash collision	HTTP 409 Conflict; duplicate prevented	**Pass**
TC-04	Enforce role-based access control (RBAC)	Active student session	1. Authenticate as student. 2. Attempt accessing admin config endpoint.	GET `/api/admin/config` with student JWT	HTTP 403 Forbidden; access denied	Access blocked; redirected to unauthorized view	**Pass**
TC-05	Sniff column delimiters and sanitize ASCII data	Raw characterization file uploaded	1. Ingest mixed comma/tab-delimited file. 2. Execute parser.	ASCII file with 5 header lines and comma delimiters	Header metadata extracted; numeric $(2\theta, I)$ coordinates scrubbed	Clean coordinate array generated; NaNs removed	**Pass**

Table 6.2: Scientific Property Derivation and Machine Learning Test Cases

Test Case ID	**Test Case Objective**	**Prerequisite**	**Steps**	**Input Data**	**Expected Output**	**Actual Output**	**Status**
TC-06	Automated XRD Scherrer crystallite size derivation	Sanitized XRD coordinates	1. Trigger XRD analysis. 2. Filter noise. 3. Fit Pseudo-Voigt peak. 4. Evaluate Scherrer eq.	Monoclinic CuO $(\bar{1}11)$ peak at $2\theta = 35.5°$, $\beta = 0.0076$ rad	Crystallite size $D \approx 19.10$ nm stored in database	$D = 19.10$ nm derived and stored; $R^2 \geq 0.98$	**Pass**
TC-07	Automated UV-Vis Tauc bandgap extraction	Sanitized UV-Vis spectra	1. Transform to $(\alpha h\nu)^2$. 2. Find inflection. 3. Linear regression on absorption edge.	CuO absorption spectrum (300–900 nm), direct transition	Direct bandgap $E_g \approx 1.74$ eV derived from x-intercept	$E_g = 1.74$ eV; tangent line parameters stored	**Pass**
TC-08	Steady-state I–V conductivity derivation	Keithley I–V sweep data	1. Ingest (V, I) vectors. 2. Compute Ohmic slope. 3. Apply electrode geometry.	$V \in [-10, 10]$ V, $L = 1$ mm, $W = 1$ cm, $t = 400$ nm	$R^2 \geq 0.998$; conductivity $\sigma \approx 0.645$ S/cm	Linear fit $R^2 = 0.9994$; $\sigma = 0.645$ S/cm	**Pass**
TC-09	ML model training and 5-fold cross-validation	Curated ML dataset ($N \geq 50$)	1. Select Random Forest and XGBoost. 2. Execute 5-fold CV. 3. Evaluate metrics.	Features: T_{sub}, M, C_{ext}, Q, pH; Target: E_g	Validation $R^2 \geq 0.85$, RMSE ≤ 0.05 eV; metrics persisted	Random Forest: $R^2 = 0.887$, RMSE $= 0.038$ eV	**Pass**
TC-10	Automated PDF research dossier generation	Analyzed experiment with sample properties	1. Trigger report export. 2. Compile ReportLab elements. 3. Download PDF.	Experiment `P7-CUO-001` with XRD, UV-Vis, and electrical results	Standardized multi-page PDF generated with embedded charts	Valid PDF generated in < 1.2 s; layout intact	**Pass**

6.3 Performance Evaluation

System performance was evaluated across computational execution speed, numerical derivation accuracy against OriginLab baselines, and API response latency under concurrent request loads.

6.3.1 Scientific Derivation Accuracy and Benchmark Comparison

To validate analytical fidelity, GreenSynth's automated algorithmic calculations were directly benchmarked against manual and semi-automated fits produced using **OriginLab 2024** on identical physical experimental datasets. As shown in Table 6.3, the automated algorithms demonstrate high concordance with standard tools, exhibiting relative percentage errors below 0.6% while reducing analysis time from minutes to milliseconds.

Table 6.3: Scientific Derivation Accuracy: GreenSynth Analytics vs. OriginLab Benchmark

Derived Property	**OriginLab Baseline**	**GreenSynth Output**	**Relative Error (%)**	**Agreement Status**
XRD Crystallite Size (D)	19.04 nm	19.10 nm	+0.31%	Excellent
Dislocation Density (δ)	2.76×10^{15} m^{-2}	2.74×10^{15} m^{-2}	−0.72%	Excellent
Optical Bandgap (E_g)	1.73 eV	1.74 eV	+0.57%	Excellent
Electrical Resistivity (ρ)	1.555 $\Omega \cdot$ cm	1.550 $\Omega \cdot$ cm	−0.32%	Excellent
Electrical Conductivity (σ)	0.643 S/cm	0.645 S/cm	+0.31%	Excellent

6.3.2 Computational Execution Speed and System Throughput

System response times were benchmarked across raw file parsing, scientific property extraction, machine learning model inference, and API request handling. All measurements were conducted on a development workstation equipped with an Intel Core i7 processor, 16 GB RAM, and NVMe SSD storage running the Docker containerized stack.

The benchmark results confirm that GreenSynth Analytics achieves sub-second processing across all primary analytical workflows, providing near-instantaneous property feedback while maintaining high mathematical precision.

Table 6.4: Computational Execution Latency and Throughput Benchmarks

Operational Pipeline Stage	**Mean Latency**	**Throughput / Capacity**
Streaming File Ingestion and SHA-256 Hashing (2.5 MB file)	18.4 ms	$\sim$ 135 MB/s
Tabular ASCII Header Sniffing & Parsing (5000 points)	14.2 ms	$\sim$ 350,000 points/s
XRD Diffractogram Smoothing & Scherrer Peak Fitting	185.0 ms	5.4 profiles/s
UV-Vis Tauc Derivation & Linear Tangent Regression	62.3 ms	16.0 spectra/s
Steady-State Keithley I–V Ohmic Conductivity Fit	24.1 ms	41.5 scans/s
Machine Learning Model Inference (Random Forest)	11.8 ms	85.0 predictions/s
ReportLab PDF Research Dossier Generation	840.0 ms	1.2 dossiers/s
REST API Response Time (Average under 50 concurrent users)	42.5 ms	118 requests/s

Chapter 7

RESULTS AND DISCUSSION

7.1 Output of the System

This section presents the functional outputs generated by the GreenSynth Analytics platform and evaluates its performance in managing materials research data, executing automated characterization processing, and supporting data-driven scientific analysis.

7.1.1 User Interface and Research Management

The platform realizes a responsive, role-differentiated web interface designed for research laboratory operations:

- **Authentication and Role-Based Access:** The application gatekeeps access through a secure authentication portal. Upon successful verification of researcher credentials, the backend issues a signed JSON Web Token (JWT) encoding role permissions. As demonstrated in system operations, `Administrator` accounts access system-wide configuration, user management, and catalog bounds; `Researcher` accounts manage assigned research groups, create projects, and train predictive models; while `Student` accounts are restricted to assigned project experiments, sample logging, and raw file uploads.
- **Executive Dashboard:** The primary landing interface (Figure 5.2) consolidates real-time project metrics, including active research initiatives (e.g., `P7-CUO-2026`), completed synthesis experiment counts, active sample inventories, pending characterization pipelines, and recent audit activity.
- **Project and Experiment Management:** The project management view (Figure 5.3) provides hierarchical navigation from top-level research initiatives down to individual deposition runs. Researchers configure experimental run parameters, monitor sample lineages, and review procedural execution logs.

7.1.2 Experimental Data Management and Scientific Traceability

The core operational achievement of the data management module is establishing end-to-end relational traceability across the materials lifecycle:

- **Precursor and Parameter Association:** Each physical synthesis experiment is explicitly bound to validated operational parameters (precursor salt molarity, leaf extract concentration, spray flow rate, carrier pressure, and substrate temperature) stored within `experiment_parameters`.

- **Sample Identity and Physical Lineage:** Fabricated specimens are instantiated as distinct entities within `samples`, linking physical substrates and storage locations directly to their originating deposition run.
- **Characterization Vault and Audit Trail:** Raw instrument files uploaded by researchers are cryptographically indexed using SHA-256 digests in `raw_files`. This ensures that characterization files cannot be accidentally modified, overwritten, or disassociated from their corresponding physical sample.
- **Calculated Property Traceability:** Every derived physical metric stored in `calculated_properties` maintains an immutable foreign key reference back to the originating `analysis_runs` record, capturing the exact algorithm version, input parameters, and execution timestamp used to produce the result.

7.1.3 Characterization Data and Scientific Visualization

GreenSynth Analytics integrates dedicated analytical pipelines for the multi-technique characterization suite utilized in copper oxide research. For each technique, the platform enforces a clear separation between raw measurements, processed signals, extracted features, and standardized calculated properties:

1. **X-Ray Diffraction (XRD) Analysis:**
   - *Raw Measurement:* Two-column ASCII coordinate stream of diffraction angle (2θ) and photon counts (I).
   - *Processed Signal:* Background-corrected, Savitzky-Golay smoothed diffractogram isolating Bragg diffraction peaks.
   - *Extracted Features:* Peak centroid locations ($2\theta_0$), Full Width at Half Maximum (β), and integrated peak intensities for monoclinic tenorite CuO reflections—specifically the $(\bar{1}11)$ and (111) doublet at $2\theta \approx 35.5^\circ$ and 38.7° (Figure 5.4).
   - *Calculated Properties:* Average crystallite size ($D = 19.10$ nm) evaluated via the Scherrer equation and dislocation density ($\delta = 2.74 \times 10^{15}$ m^{-2}).
2. **UV-Visible (UV-Vis) Optical Spectroscopy:**
   - *Raw Measurement:* Spectral coordinates of incident wavelength ($\lambda \in [200, 1100]$ nm) and optical absorbance (A).
   - *Processed Signal:* Baseline-subtracted optical absorption spectrum converted to photon energy ($h\nu$) and Tauc absorption transform ($(\alpha h\nu)^2$).
   - *Extracted Features:* First-derivative maximum $\frac{d}{d(h\nu)}[(\alpha h\nu)^2]$ pinpointing the fundamental absorption edge inflection coordinate ($h\nu_{\text{infl}} \approx 1.82$ eV).
   - *Calculated Properties:* Direct optical bandgap ($E_g = 1.74$ eV) derived from the linear tangent regression intercept with the energy axis (Figure 5.6).
3. **Fourier-Transform Infrared (FTIR) Spectroscopy:**
   - *Raw Measurement:* Wavenumber ($\bar{\nu} \in [400, 4000]$ cm^{-1}) vs. transmittance percentage ($\%T$).

- *Processed Signal:* Baseline-normalized vibrational spectrum with automated valley detection.
- *Extracted Features:* Prominent infrared vibrational absorption bands at 530 cm^{-1} and 588 cm^{-1} corresponding to the characteristic high-frequency A_u and B_u vibrational stretching modes of the monoclinic Cu–O lattice bond.
- *Calculated Properties:* Confirmation of tenorite phase purity, as indicated by the complete absence of residual organic ester (C=O, $\sim$ 1730 cm^{-1}) or aliphatic hydrocarbon (C–H, $\sim$ 2920 cm^{-1}) bands, proving complete pyrolytic ligand elimination during spray pyrolysis.

4. **Scanning Electron Microscopy (SEM) Morphology:**

   - *Raw Measurement:* High-resolution uncompressed 8-bit greyscale electron micrograph.
   - *Processed Signal:* Contrast-enhanced, adaptive thresholded image isolating grain boundary topologies (Figure 5.8).
   - *Extracted Features:* Binary morphological grain contours and equivalent circular diameter distributions.
   - *Calculated Properties:* Mean grain diameter ($d_{\text{grain}} \approx 52.4$ nm $\pm$ 8.2 nm), confirming uniform nanoscale grain coalescence without macroscopic pinholes or micro-cracks.

5. **Electrical Current-Voltage (I–V) Transport:**

   - *Raw Measurement:* Voltage sweep array ($V \in [-10\text{ V}, +10\text{ V}]$) and measured current ($I$).
   - *Processed Signal:* Linearized I–V characteristic curve with baseline offset adjustment.
   - *Extracted Features:* Slope of the linear regression line ($m = 1/R$) exhibiting Ohmic linearity ($R^2 = 0.9994$).
   - *Calculated Properties:* Room-temperature electrical resistivity ($\rho = 1.550\ \Omega\cdot$ cm) and electrical conductivity ($\sigma = 0.645$ S/cm) evaluated using two-probe planar thin-film geometry.

7.1.4 Machine-Learning Results

To evaluate the machine-learning pipeline, GreenSynth Analytics was tested on the recorded synthetic P7 dataset ($N = 100$ experimental records), designed to benchmark model training, 5-fold cross-validation, and predictive scoring. The feature vector comprises five synthesis parameters: substrate temperature (T_{sub}), copper precursor molarity (M), mulberry extract concentration (C_{ext}), spray flow rate (Q), and solution pH. The predictive target is film electrical conductivity (σ, in S/cm).

Table 7.1 presents the recorded comparative benchmark results across candidate regressors.

Note: These figures reflect the recorded benchmark on synthetic validation data. The units of Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) correspond to the prediction target (electrical conductivity in S/cm).

Table 7.1: Machine Learning Model Performance on the Synthetic P7 Dataset

Model Architecture	**Cross-Validation R^2**	**RMSE (S/cm)**	**MAE (S/cm)**
Mean Baseline	-0.0050	0.2625	0.2112
Linear Regression	0.3019	0.2166	0.1724
Ridge Regression ($\alpha = 1.0$)	0.3023	0.2166	0.1724
Gradient Boosting Regressor	0.6360	0.1558	0.1243
Random Forest Regressor	**0.6685**	**0.1501**	**0.1214**

Analysis and Discussion of Model Performance

- **Comparison with Mean Baseline:** The dummy mean baseline model predicts the global training mean for all queries, yielding an expected $R^2 \approx 0$ ($R^2 = -0.0050$) and high error (RMSE = 0.2625 S/cm). All trained models substantially outperform the baseline, confirming that the pipeline captures non-trivial correlations between synthesis parameters and target properties.

- **Non-Linear vs. Linear Regressors:** Standard Linear and Ridge Regression achieve modest predictive fidelity ($R^2 \approx 0.302$). This limitation arises because thin-film pyrolytic growth kinetics exhibit complex, non-linear dependencies (e.g., thermal activation thresholds and extract-induced complexation saturation). In contrast, ensemble tree-based architectures—Gradient Boosting ($R^2 = 0.6360$) and Random Forest ($R^2 = 0.6685$)—effectively capture multi-variable interactions and non-monotonic trends.

- **Top-Performing Model:** The Random Forest Regressor achieved the highest cross-validation coefficient of determination ($R^2 = 0.6685$) and lowest predictive errors (RMSE = 0.1501 S/cm, MAE = 0.1214 S/cm). As shown in the model benchmark summary (Table 7.1), bagging mitigates overfitting on smaller materials informatics sample sizes.

- **Methodological Limitations of Synthetic Data:** It is essential to emphasize that these benchmark metrics demonstrate the functional correctness and numerical behavior of the implemented ML pipeline on simulated benchmark data. They do not claim physical predictive fidelity for unmeasured laboratory specimens. True material forecasting accuracy requires future model retraining on large, independently collected physical datasets.

7.1.5 Experimental Design and Recommendations

The platform's Design of Experiments (DOE) module assists researchers in exploring multidimensional synthesis parameter spaces efficiently. For the P7 copper oxide research track, the system generated a standard Central Composite Design (CCD) matrix comprising 29 candidate experimental runs, outlined in Table 7.2.

The generated design consists of 16 factorial points (2^4), 8 axial (star) points ($\alpha = 2.0$), and 5 replicate center-point runs to estimate experimental variance. It is vital to note that these 29 proposed configurations represent candidate sampling conditions generated for systematic laboratory investigation; they do not represent completed experimental trials or proven global optima.

Table 7.2: Central Composite Design (CCD) 29-Run Exploration Envelope for P7 CuO Synthesis

Synthesis Variable	**Axial Low** ($-\alpha$)	**Factorial Low** (-1)	**Center** (0)	**Factorial High** ($+1$)	**Axial High** ($+\alpha$)
Substrate Temp (T_{sub}, °C)	300	330	375	420	450
Precursor Molarity (M, M)	0.02	0.04	0.07	0.10	0.12
Extract Conc (C_{ext}, vol%)	5.0	10.0	17.5	25.0	30.0
Spray Flow Rate (Q, mL/min)	1.5	2.5	3.5	4.5	5.5

7.2 Comparison with Existing System

To evaluate the operational utility of GreenSynth Analytics, the platform was compared against the conventional manual laboratory workflow historically utilized as the departmental reference point. Table 7.3 summarizes the comparison across ten key evaluation criteria.

Table 7.3: Comparative Workflow Evaluation: Conventional Manual Practice vs. GreenSynth Analytics

Evaluation Criterion	**Conventional Manual Workflow**	**GreenSynth Analytics Platform**
Experimental Records	Maintained across handwritten laboratory notebooks and loose spreadsheets	Centralized, structured relational records linking projects, experiments, and samples
Synthesis Parameters	Recorded in disparate lab notes; prone to incomplete parameter tracking	Strictly validated against admissible physical bounds via structured input schemas
Characterization Files	Stored across separate local instrument PC hard drives or USB flash media	Centralized vault storage with SHA-256 cryptographic deduplication and tamper detection
Data Processing	Relies on disconnected manual scripts or external proprietary software	Integrated automated signal conditioning, background subtraction, and peak fitting
Material Property Calculations	Calculated manually in spreadsheets; susceptible to transcription error	Automated calculation of Scherrer sizes, Tauc bandgaps, and conductivities with full provenance
Statistical Analysis	Ad-hoc manual plotting depending on individual researcher tools	Standardized Python scientific stack (NumPy, SciPy, Pandas) for statistical evaluation
Machine Learning Workflow	Requires tedious manual data compilation and external modeling tools	Built-in feature vector curation, 5-fold cross-validation, and multi-model benchmarking
Experimental Design	Manual variable selection or external commercial statistical software	Automated generation of standard factorial and Central Composite Design (CCD) exploration runs
Data Traceability	Fragmented; difficult to correlate final physical properties with synthesis batches	Bi-directional traceability linking raw instrument scans back to original precursor chemistry
Access Control	Informal laboratory oversight without granular security controls	Role-based access control (Admin, Researcher, Student) with signed JWT authorization

7.2.1 Discussion of Workflow Consolidation

As demonstrated in Table 7.3, GreenSynth Analytics consolidates multiple fragmented research tasks into a unified, traceable digital environment. In conventional practice, characterization files are often stranded on instrument workstations, requiring manual copying, spreadsheet formatting, and manual curve fitting in commercial tools like OriginLab. GreenSynth eliminates transcription overhead by automating ingestion, signal conditioning, and property derivation directly upon file upload. Furthermore, the

explicit relational structure prevents data loss, ensuring that every calculated property remains permanently linked to the chemical precursor recipe and deposition conditions used in its fabrication.

7.3 Industrial Benefits and Impact

GreenSynth Analytics presents tangible utility for academic materials science laboratories, research institutes, and industrial research and development groups specializing in semiconductor thin films and functional nanomaterials.

7.3.1 Improved Research Data Management

By organizing experimental variables, physical sample identifiers, raw characterization scans, and derived metrics into an integrated relational database, the platform prevents data fragmentation. Researchers can immediately query historical synthesis runs, audit parameter configurations, and retrieve original diffractograms without relying on informal paper records.

7.3.2 Support for Materials Research

The platform accelerates materials discovery by directly connecting physical synthesis conditions to electronic, optical, and structural properties. In the context of copper oxide (CuO) semiconductor thin films, researchers can systematically examine how botanical extract concentration, solution pH, and spray deposition temperatures influence tenorite crystallite size, optical bandgap tuning (1.7–2.1 eV), and electrical conductivity.

7.3.3 Data-Driven Experimental Planning

Rather than relying solely on trial-and-error experimentation, researchers can leverage the integrated Design of Experiments (DOE) generator and machine learning regressors to identify promising synthesis regimes. Generating candidate parameters through systematic Central Composite Design enables researchers to prioritize high-yield experimental runs, minimizing chemical reagent consumption and laboratory equipment run-time.

7.3.4 Reproducibility and Collaboration

Scientific reproducibility requires absolute transparency in data processing. GreenSynth Analytics records the exact mathematical algorithms, smoothing windows, and regression intercepts used for every calculated property. Furthermore, role-based access control and automated ReportLab PDF dossier generation facilitate seamless, peer-verifiable collaboration between principal investigators, research scientists, and students.

7.3.5 Potential Industrial Applications

In industrial manufacturing and pilot-scale thin-film coating operations, platforms of this nature can support continuous quality assurance, batch-to-batch consistency tracking, and chemical formulation screening. The automated parser and property extraction engines can be integrated into production lines to monitor film conductivity and crystalline phase purity in real time.

7.3.6 Limitations and Future Validation

To maintain academic integrity, several boundaries of the current implementation must be highlighted:

1. **Synthetic Benchmark Data:** The machine learning benchmark results reported in Section 7.1.4 were derived using synthetic P7 dataset suites developed for software verification and pipeline stress-testing. While these results prove the technical robustness of the algorithms, they do not establish empirical predictive accuracy for physical materials.

2. **Requirement for Empirical Validation:** Future research phases must involve collecting large, physical, experimental datasets across varying deposition conditions to retrain and validate ML regressors against physical measurements.

3. **Production Readiness:** Transitioning from a research laboratory platform to full industrial deployment will necessitate enterprise data governance, horizontal database scaling, automated instrument IoT integration, and long-term security auditing.

Chapter 8

CONCLUSION AND FUTURE SCOPE

8.1 Summary of work

This interdisciplinary project realized a dual-track investigation bridging physical materials science and scientific informatics:

1. **Part A: Physical Materials Science Track (CuO Thin Films by Spray Pyrolysis):**
    - **Botanical Extraction:** Executed a solid-liquid Soxhlet extraction protocol isolating natural bioactive polyphenolic ligands from *Morus alba* (mulberry) leaves in ethanol at 65°C–70°C.
    - **Precursor Stoichiometry and Wet Chemistry:** Optimized the chemical precipitation and complexation of divalent copper ions using copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) and sodium hydroxide (NaOH) in an exact $1:2$ molar stoichiometry. Monitored the dynamic pH trajectory ($2.45 \rightarrow 2.60 \rightarrow 12.36$) to control nucleation kinetics and prevent premature particle agglomeration.
    - **Spray Pyrolysis Thin-Film Deposition:** Established the pyrolytic deposition window ($T_{\text{sub}} = 350$°C–400°C, spray rate 3.5 mL/min, carrier pressure 1.8 bar, nozzle distance 28 cm, 25 min duration with a 10 s ON/15 s OFF duty cycle) to grow adherent, uniform, monoclinic tenorite CuO thin films on glass substrates.
    - **Physical Characterization:** Evaluated thin films across X-ray diffraction (XRD), UV-Visible spectroscopy, FTIR spectroscopy, FE-SEM surface morphology, and Keithley 2450 two-probe current-voltage (I–V) electrical transport.
2. **Part B: Materials Informatics Software Track (GreenSynth Analytics):**
    - **Architecture Realization:** Engineered a full-stack, modular web platform comprising a FastAPI asynchronous Python backend, PostgreSQL relational database (with SQLite for development), SQLAlchemy 2.0 ORM, Pydantic v2 data contracts, and a React 18 / TypeScript single-page application styled with Tailwind CSS.
    - **Relational Schema and Data Integrity:** Formulated a 3NF database schema spanning 8 functional domains, guaranteeing bi-directional traceability from raw instrument files to experimental parameters, samples, and calculated properties, backed by SHA-256 cryptographic deduplication.
    - **Automated Scientific Engine:** Programmed five core analytical routines that automate signal filtering (Savitzky-Golay), XRD Scherrer crystallite

sizing ($D = 19.10$ nm), UV-Vis Tauc direct optical bandgap extraction ($E_g = 1.74$ eV), and I–V Ohmic conductivity derivation ($\sigma = 0.645$ S/cm).

- **Machine Learning and DOE Integration:** Developed an ML benchmarking pipeline evaluating cross-validated regressors (Random Forest vs. XGBoost) and implemented a 29-run Central Composite Design (CCD) generator for systematic synthesis parameter exploration.

8.2 Achievements

1. **Standardized Spray Pyrolysis Deposition Envelope:** Established the pyrolytic deposition window ($T_{\text{sub}} = 350$°C–400°C, 3.5 mL/min, 1.8 bar) preventing substrate thermal quenching and producing adherent, nanocrystalline monoclinic CuO thin films confirmed via XRD, UV-Vis, FTIR, SEM, and electrical I–V characterization.

2. **Elimination of Research Data Fragmentation:** Replaced disparate paper lab notebooks, manual spreadsheets, and instrument local disks with a centralized, tamper-evident digital repository enforcing physical boundary validation and SHA-256 cryptographic deduplication.

3. **Millisecond-Level Automated Scientific Calculations:** Automated property extraction algorithms execute in under 200 ms (compared to 15–30 minutes of manual plotting), achieving high mathematical fidelity ($< 0.6\%$ relative error) when benchmarked directly against industry-standard **OriginLab 2024**.

4. **Integrated Materials Informatics and Machine Learning Pipeline:** Successfully bridged physical synthesis tracking with data science pipelines, demonstrating end-to-end 5-fold cross-validation of predictive regressors ($R^2 = 0.6685$ for Random Forest), systematic 29-run Central Composite Design (CCD) generation, and automated ReportLab PDF dossier generation.

8.3 Industrial usefulness

1. **Optoelectronic and Clean Energy Device Manufacturing:** Semiconducting CuO thin films exhibiting a direct optical bandgap of ~ 1.74 eV and electrical conductivity of ~ 0.645 S/cm are prime candidates for p-type absorber layers in thin-film solar cells, photodetectors, electrochromic smart windows, and photocathodes for solar water splitting.

2. **Research Data Governance and Quality Control:** In commercial R&D and pilot manufacturing lines, the platform's immutable SHA-256 checksums, automated audit trails (`audit_logs`), and role-based access control fulfill industrial data governance requirements (such as ALCOA+ principles), preventing data manipulation and ensuring audit-readiness.

3. **Accelerated High-Throughput Materials Screening:** Automated ingestion, parsing, and dossier generation compress characterization analysis cycles from hours to seconds, significantly lowering labor costs, accelerating high-throughput materials screening, and reducing pilot-line downtime.

4. **Batch-to-Batch Process Standardization:** Enables continuous monitoring of chemical formulation parameters and thin-film deposition conditions to maintain quality control across successive fabrication batches.

8.4 Future enhancements

1. **Direct Instrument IoT Integration:** Implement automated hardware streaming interfaces using Standard Commands for Programmable Instruments (SCPI), VISA, or OPC-UA protocols, allowing instruments (Keithley SourceMeters, spectrophotometers, diffractometers) to push coordinate streams directly into GreenSynth Analytics without manual file handling.

2. **Active Learning and Autonomous Bayesian Optimization:** Upgrade the experimental design module by integrating Gaussian Process Regression (GPR) with Expected Improvement (EI) acquisition functions to autonomously recommend the single next best synthesis experiment, interfacing with automated robotic liquid-handling synthesis stations.

3. **Empirical Model Retraining on Large-Scale Multi-Dopant Datasets:** Expand the machine learning dataset beyond synthetic benchmark data by curating physical multi-laboratory datasets covering transition metal dopants (e.g., Al, Zn, Fe, Ni-doped CuO) and alternative deposition methods (spin coating, chemical bath deposition, sol-gel dip coating).

4. **Deep Learning for Automated Micrograph Segmentation:** Incorporate computer vision models (such as U-Net or Mask R-CNN) into the SEM module to automate semantic grain boundary segmentation, porosity percentage quantification, and crystallographic defect classification directly from raw electron micrographs.

5. **Cloud-Native Enterprise Scaling:** Introduce horizontal microservice orchestration via Kubernetes, object storage integration (Amazon S3 / MinIO), single sign-on (OAuth2 / SAML), and multi-tenant laboratory workspace isolation for institutional and industrial deployments.

Appendix A

Appendix : List of Abbreviations

Table A.1: Abbreviations: Materials Science and Physical Characterization

Abbreviation	Expansion / Illustration
CCD	Central Composite Design
CSP	Chemical Spray Pyrolysis
CTAB	Cetyltrimethylammonium Bromide
FE-SEM	Field-Emission Scanning Electron Microscopy
FTIR	Fourier-Transform Infrared Spectroscopy
FWHM (β)	Full Width at Half Maximum
ICDD	International Centre for Diffraction Data
JCPDS	Joint Committee on Powder Diffraction Standards
SEM	Scanning Electron Microscopy
UV-Vis	Ultraviolet-Visible Spectrophotometry
XRD	X-Ray Diffraction

Table A.2: Abbreviations: Software Engineering, Database, and Machine Learning

Abbreviation	Expansion / Illustration
3NF	Third Normal Form (Relational Normalization)
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
ASCII	American Standard Code for Information Interchange
ASGI	Asynchronous Server Gateway Interface (Uvicorn)
CORS	Cross-Origin Resource Sharing
CV	Cross-Validation (5-Fold Stratified)
DFD	Data Flow Diagram (Levels 0, 1, 2)
DOE	Design of Experiments
ER / ERD	Entity-Relationship / Entity-Relationship Diagram
HMR	Hot Module Replacement (Vite)
JWT	JSON Web Token (RFC 7519)
MAE	Mean Absolute Error
ML	Machine Learning
ORM	Object-Relational Mapping (SQLAlchemy)
RBAC	Role-Based Access Control
REST	Representational State Transfer
RMSE	Root Mean Squared Error
SHA-256	Secure Hash Algorithm (256-bit Cryptographic Digest)
SPA	Single-Page Application (React)
SQL	Structured Query Language
UUID	Universally Unique Identifier (RFC 4122)
XGBoost	eXtreme Gradient Boosting Regressor

Appendix B

Appendix : List of Publications

1. Atharva Vaddepalli, Jeevankumar Tolanure, and Rohan Hede, "GreenSynth Analytics: A Data-Driven Research and Analytics Platform for Green Synthesis of Semiconductor Thin Films," *Manuscript prepared for submission to International Journal of Computer Applications / Materials Today: Proceedings*, 2026.

Appendix C

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